

Design and Analysis of a Closed-Loop Buck Converter Using PI Control

1. Introduction

1.1 Background

A buck converter is a DC–DC power converter that steps down a higher DC voltage to a lower DC voltage while maintaining high conversion efficiency. Unlike linear voltage regulators, which dissipate excess power as heat, a buck converter uses high-frequency switching elements together with an inductor and capacitor to efficiently transfer energy from the input to the output.

Buck converters are widely used in modern electronic systems where different components require different voltage levels from the same power source. Typical applications include laptop power supplies, battery-powered devices, electric vehicles, communication systems, industrial power supplies, and embedded electronic circuits.

For example, a laptop charger converts the AC supply from the electrical outlet into a DC voltage of approximately 19 V. However, the internal electronic components of the laptop do not all operate at this voltage. The processor, memory modules, and other digital circuits require much lower regulated voltages, typically around 5 V, 3.3 V, or even lower. Therefore, a buck converter is employed inside the system to step down the higher DC voltage to the required operating voltages with high efficiency.

1.2 Need for Voltage Regulation

Although a buck converter can reduce the input voltage to a lower output voltage, practical operating conditions are rarely constant. The input supply voltage may vary due to fluctuations in the source or battery discharge. Similarly, the connected load may change as different electronic devices are switched on or off.

These variations disturb the output voltage of the converter. However, many electronic systems require a constant and well-regulated supply voltage for reliable and safe operation. Even small deviations from the desired voltage may reduce performance or damage sensitive electronic components.

Therefore, it is not sufficient for a converter to simply reduce voltage. It must also regulate the output voltage despite disturbances caused by changes in the input supply or load conditions. This requirement is known as voltage regulation.

1.3 Closed-Loop Buck Converter

The simplest method of operating a buck converter is open-loop control, where the duty ratio is fixed. In this case, the output voltage depends directly on the operating conditions. Any change in input voltage or load resistance causes the output voltage to deviate from its desired value.

To overcome this limitation, a closed-loop control system is employed. In a closed-loop buck converter, the actual output voltage is continuously measured and compared with the desired reference voltage. The difference between these two quantities is known as the error signal.

This error signal is processed by a controller, which automatically adjusts the switching duty ratio of the buck converter. By continuously modifying the duty ratio, the controller maintains the output voltage close to its desired value even when the input voltage, load, or reference voltage changes.

The overall closed-loop system therefore provides improved voltage regulation, disturbance rejection, and reference tracking compared with an open-loop converter.

1.4 Project Objectives

The objective of this project is to design and analyse a closed-loop buck converter capable of maintaining a regulated output voltage under varying operating conditions.

The work begins with the design of the power stage by selecting suitable values of the inductor and capacitor based on converter specifications. The converter operation is first verified through open-loop simulation in LTspice.

A mathematical model of the converter is then developed using averaged modelling and small-signal analysis to obtain the control-to-output transfer function. Based on this model, a PI controller is designed using frequency-domain techniques to satisfy the desired stability and dynamic performance requirements.

Finally, the complete closed-loop system is implemented in MATLAB/Simulink and validated under various operating conditions, including reference voltage changes, input voltage variations, and load resistance variations.

2. Buck Converter Design

The first step in the development of the proposed closed-loop buck converter is the design of the power stage. This involves selecting the converter specifications and calculating the values of the passive components required to achieve the desired performance. The converter is designed to operate in Continuous Conduction Mode (CCM), and the component values are selected based on the specified current and voltage ripple limits.

2.1 Operating Principle

A buck converter is a switched-mode DC–DC converter used to obtain a lower DC voltage from a higher DC input voltage. The basic converter consists of a MOSFET switch, a diode, an inductor, a capacitor and a load resistance.

The operation of the converter can be divided into two switching intervals.

ON State

When the MOSFET is turned ON, the input voltage is directly applied across the inductor and load. The diode becomes reverse biased and does not conduct. During this interval, the inductor stores energy while its current increases approximately linearly. The capacitor supplies the load together with the inductor and helps maintain a smooth output voltage.

OFF State

When the MOSFET is turned OFF, the input supply is disconnected from the converter. Since the current through an inductor cannot change instantaneously, the inductor releases its stored energy through the diode to the load and capacitor. Consequently, the inductor current decreases approximately linearly until the next switching cycle begins.

By continuously repeating these two operating states at a high switching frequency, the converter produces a regulated average output voltage. Under ideal continuous conduction mode (CCM) operation, the output voltage is related to the input voltage by

$$V_o = DV_{in}$$

where D is the duty ratio.

2.2 Design Specifications

The proposed buck converter is designed to convert a 24 V DC input into a regulated 12 V DC output while supplying a rated load current of 2 A. The converter operates in Continuous Conduction Mode (CCM) with a switching frequency of 100 kHz.

Under ideal CCM operation, the output voltage of a buck converter is given by

$$V_o = DV_{in}$$

where D is the duty ratio.

Therefore,

$$D = \frac{V_o}{V_{in}} = \frac{12}{24} = 0.5$$

Hence, the converter requires a duty ratio of 50%.

Inductor Design

The inductor is selected based on the allowable inductor current ripple. A ripple current that is too small requires a larger inductance, increasing the size, cost, and weight of the converter. Conversely, a very large ripple increases the RMS current, conduction losses, and output current ripple. Therefore, a ripple current equal to 20% of the rated output current is selected as a practical design compromise.

Thus,

$$\Delta I_L = 0.2I_o$$

$$\Delta I_L = 0.2 \times 2 = 0.4A$$

For a buck converter operating in CCM, the required inductance is

$$L = \frac{(V_{in} - V_o)D}{f_s \Delta I_L}$$

Substituting the design values,

$$L = \frac{(24 - 12) \times 0.5}{100 \times 10^3 \times 0.4}$$

$$L = 1.5 \times 10^{-4} H$$

$$L = 150 \mu H$$

Capacitor Design

The output capacitor is selected based on the allowable output voltage ripple. A smaller voltage ripple requires a larger capacitor, which increases cost and physical size. For this project, the output voltage ripple is limited to 0.5% of the rated output voltage.

Therefore,

$$\begin{aligned}\Delta V_o &= 0.005V_o \\ \Delta V_o &= 0.005 \times 12 = 60\text{mV}\end{aligned}$$

Assuming negligible capacitor ESR, the required capacitance in CCM is

$$C = \frac{\Delta I_L}{8f_s \Delta V_o}$$

Substituting the design values,

$$\begin{aligned}C &= \frac{0.4}{8 \times 100 \times 10^3 \times 0.06} \\ C &= 8.33 \times 10^{-6} \text{ F} \\ C &= 8.33 \mu\text{F}\end{aligned}$$

Since 8.33 μF is not a standard capacitor value, the nearest higher standard value of 10 μF is selected for the converter.

The final component values used throughout the project are summarized in Table 2.1.

Table 2.1 Final Design Specifications of the Buck Converter

Parameter	Value
Input Voltage	24 V
Output Voltage	12 V
Rated Output Current	2 A
Duty Ratio	0.5
Switching Frequency	100 kHz
Inductor	150 μH
Capacitor	10 μF

2.3 Open-Loop Verification in MATLAB/Simulink

After selecting the converter parameters, the designed buck converter was implemented in MATLAB/Simulink to verify the analytical design. The objective of the open-loop simulation was to confirm that the calculated inductor and capacitor values produce the desired output voltage while satisfying the ripple specifications before proceeding to the small-signal modelling and controller design.

The converter was operated with a fixed duty ratio of 0.5, corresponding to the theoretical duty cycle required to obtain a nominal output voltage of 12 V from a 24 V input source. Since the converter operates in open loop, no feedback controller was employed and the duty ratio remained constant throughout the simulation.

The complete Simulink model used for the open-loop verification is shown in Figure 2.1. The model consists of a DC voltage source, a MOSFET switch, a freewheeling diode, an inductor, an output capacitor and a resistive load. A PWM generator based on the comparison of a high-frequency triangular carrier signal with a constant duty-cycle reference was used to generate the switching pulses for the MOSFET. The switching frequency was maintained at 100 kHz throughout the simulation.

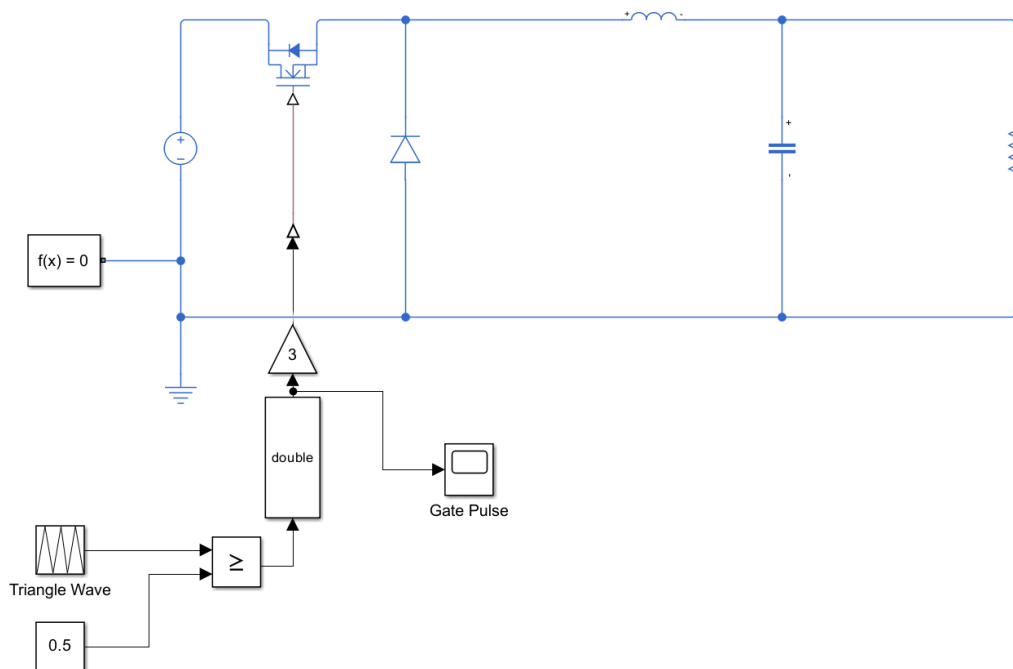


Figure 2.1 Open-loop Buck Converter Model Developed in MATLAB/Simulink

The PWM switching pulse generated by the comparator is shown in Figure 2.2. The pulse train exhibits a duty ratio of approximately 50%, thereby controlling the conduction interval of the MOSFET.

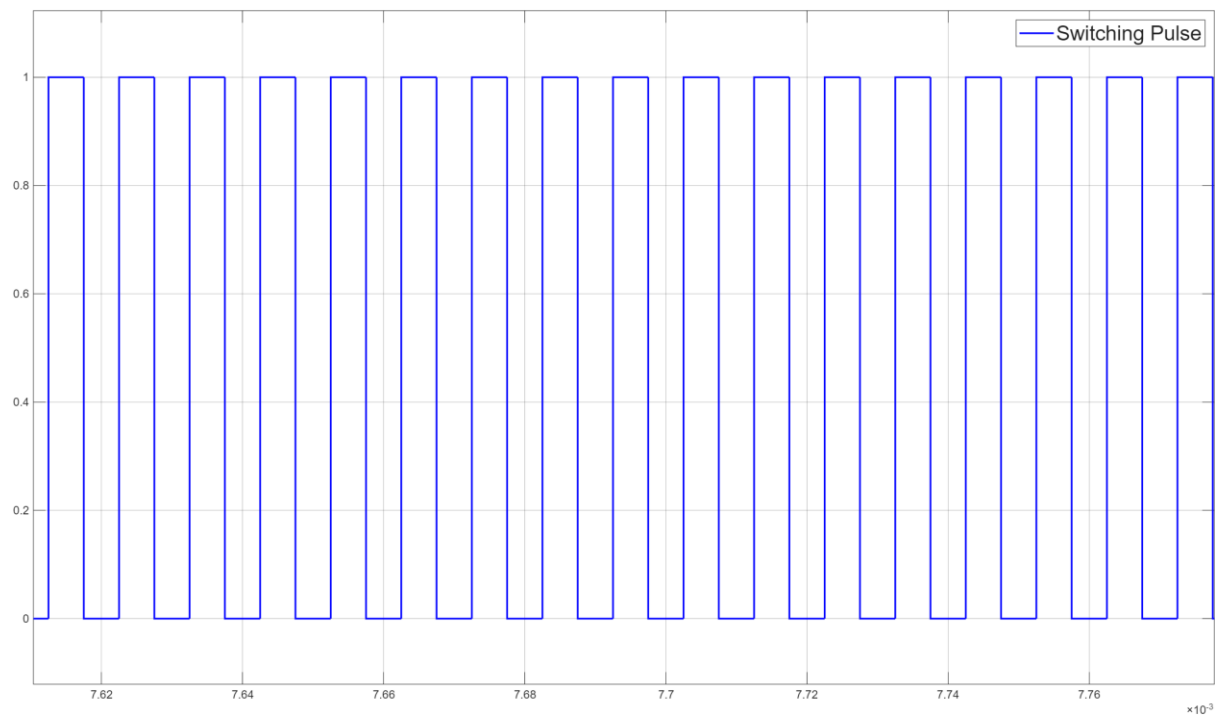


Figure 2.2 PWM Switching Pulse

The transient response of the output voltage is illustrated in Figure 2.3. As observed, the converter initially exhibits an overshoot before gradually settling to its steady-state value due to the energy exchange between the inductor and capacitor. After the transient period, the output voltage stabilizes close to the desired value of 12 V.

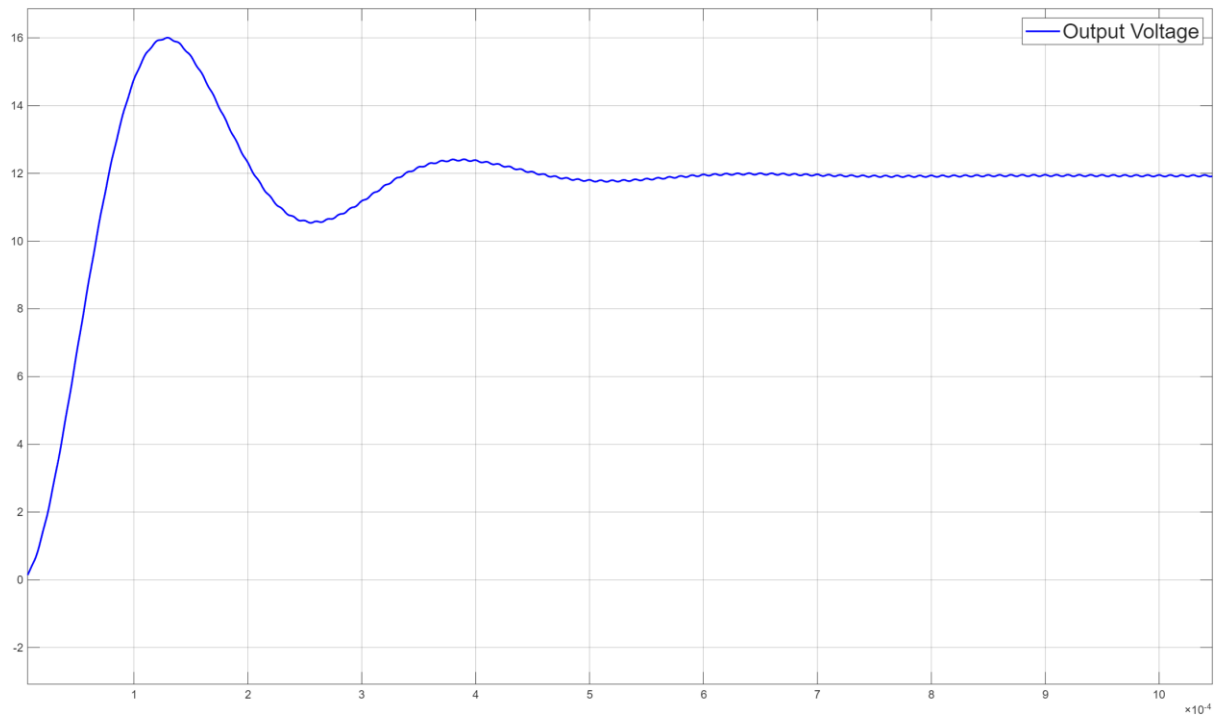


Figure 2.3 Output Voltage During Start-up

After the converter reached steady-state operation, the output voltage and inductor current were observed over several switching cycles to evaluate the ripple characteristics. The steady-state output voltage waveform is shown in Figure 2.4, while the corresponding inductor current waveform is shown in Figure 2.5.

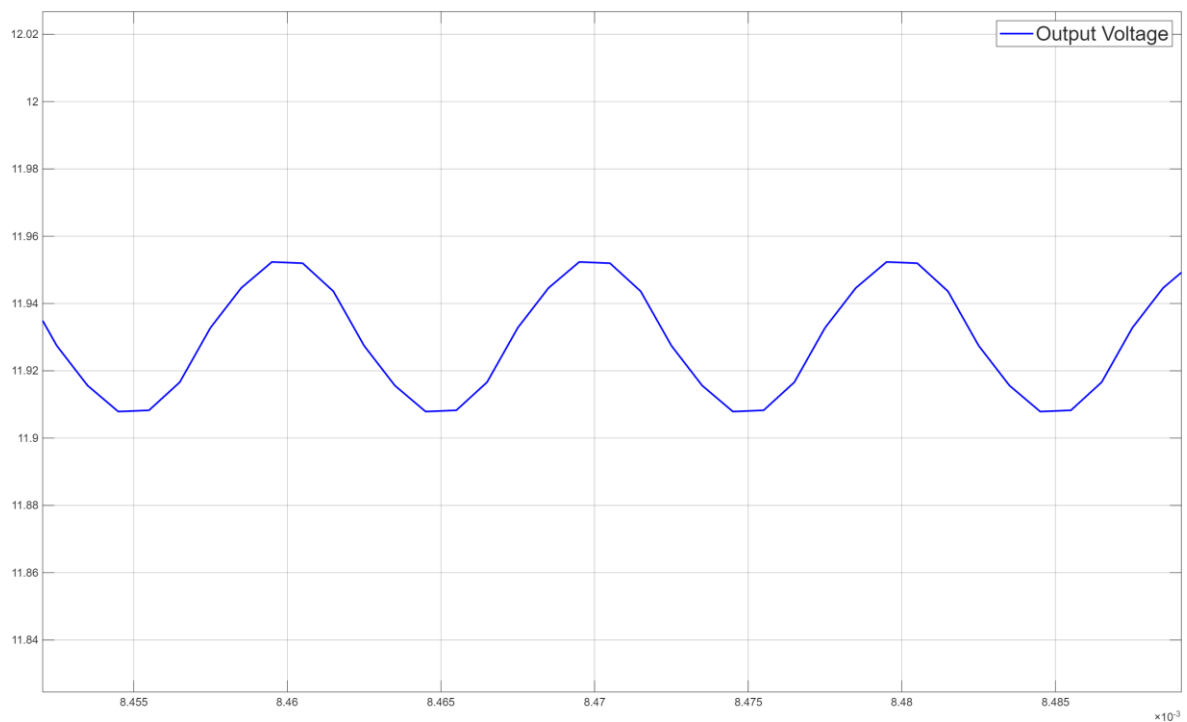


Figure 2.4 Steady-State Output Voltage

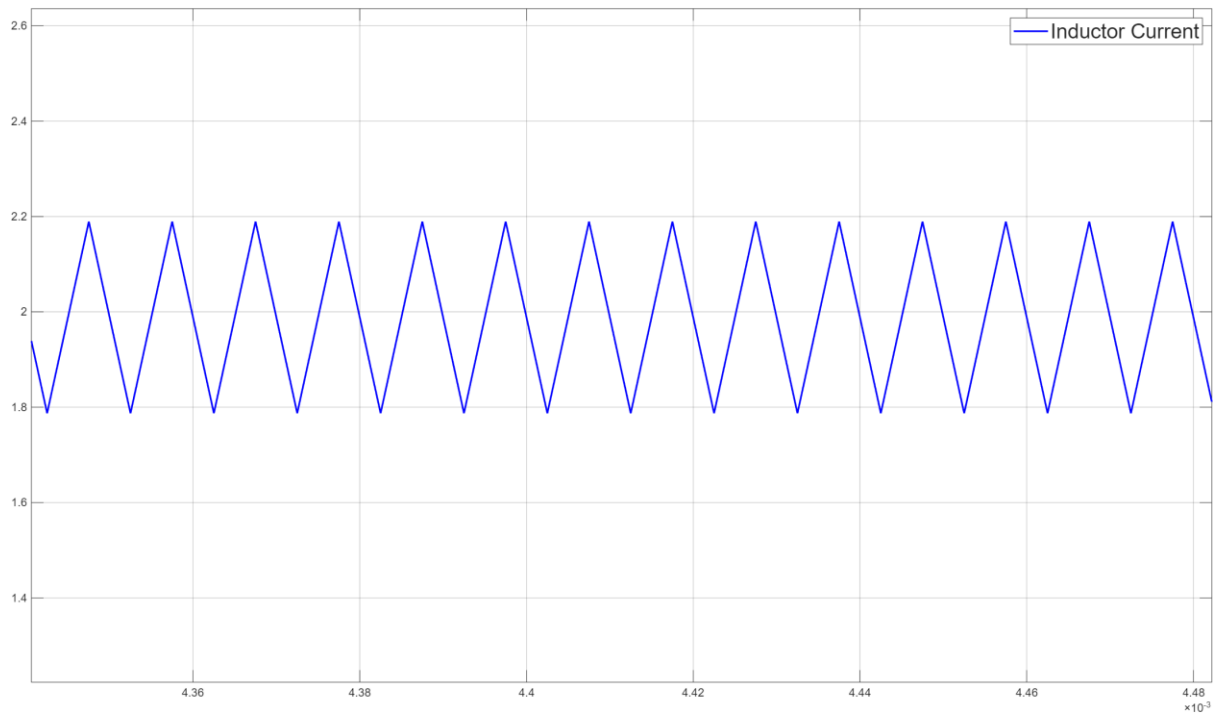


Figure 2.5 Steady-State Inductor Current

The simulation waveforms were exported to the MATLAB workspace for further analysis. A MATLAB script was developed to automatically compute the average values, RMS values and peak-to-peak ripple of the converter waveforms. The complete MATLAB analysis script used for obtaining these results is provided in Appendix A, while the simulation parameters and solver configuration are included in Appendix B.

The steady-state performance obtained from the simulation is summarized in Table 2.2.

Table 2.2. Steady-State Performance of the Open-Loop Buck Converter

Parameter	Value
Average Output Voltage	11.9301 V
Average Output Current	1.9884 A
Average Inductor Current	1.9884 A
Output Voltage Ripple (Peak-to-Peak)	45.27 mV
Inductor Current Ripple (Peak-to-Peak)	0.4022 A
Output Voltage RMS	11.9301 V
Output Current RMS	1.9884 A
Inductor Current RMS	1.9931 A

The simulation results closely agree with the theoretical calculations presented in the previous section. The average output voltage is approximately 12 V, while the measured inductor current ripple of 0.4022 A closely matches the design value of

0.4 A. Similarly, the measured output voltage ripple of 45.27 mV is lower than the specified design limit of 60 mV. These results validate the analytical design procedure and confirm that the selected inductor and capacitor values satisfy the required performance specifications. Consequently, the open-loop buck converter model provides a suitable foundation for the development of the small-signal model and the subsequent closed-loop controller design.

2.4 Open-Loop Validation in Ltspice

To further validate the converter design, the open-loop buck converter was also implemented in LTspice. Unlike MATLAB/Simulink, LTspice is a SPICE-based circuit simulator that performs detailed circuit-level simulation using semiconductor device models. The purpose of this simulation was to verify the switching behaviour of the converter using realistic MOSFET and diode characteristics.

The LTspice implementation of the converter is shown in Figure 2.6. The circuit consists of a 24 V DC input source, a MOSFET switch, a Schottky freewheeling diode, a 150 μ H inductor, a 10 μ F output capacitor and a 6 Ω resistive load. A pulse voltage source operating at a switching frequency of 100 kHz with a duty ratio of 50% was used to drive the MOSFET gate.

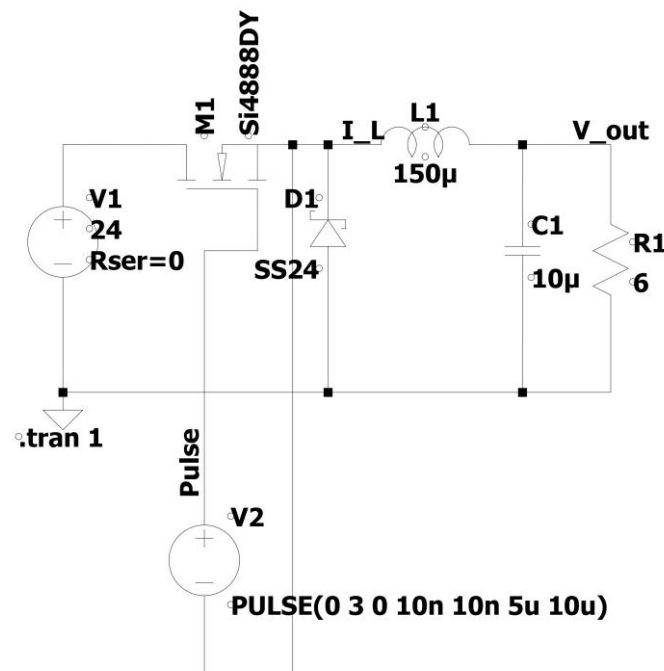


Figure 2.6 Open-Loop Buck Converter Implemented in LTspice

Manufacturer device models were employed for both the MOSFET and the freewheeling diode to represent practical switching characteristics.

The steady-state output voltage waveform obtained from LTspice is presented in Figure 2.7. The waveform shows that the converter produces an output voltage close to the desired value of 12 V with a small ripple superimposed due to the switching action.

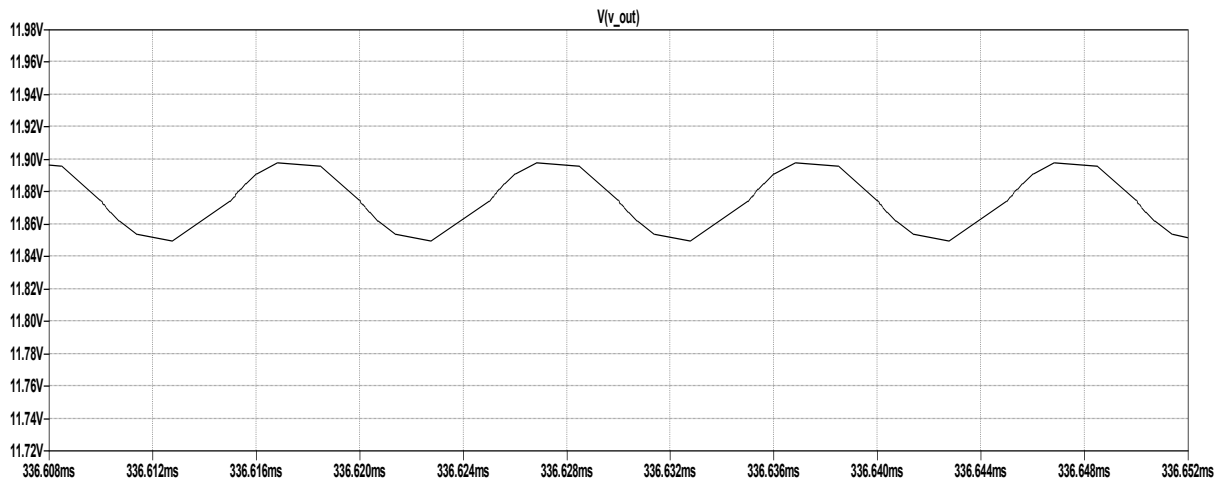


Figure 2.7 Steady-State Output Voltage Waveform Obtained from LTspice

The corresponding steady-state inductor current is shown in Figure 2.8. As expected for continuous conduction mode (CCM), the inductor current remains continuous throughout the switching cycle and exhibits the characteristic triangular ripple.

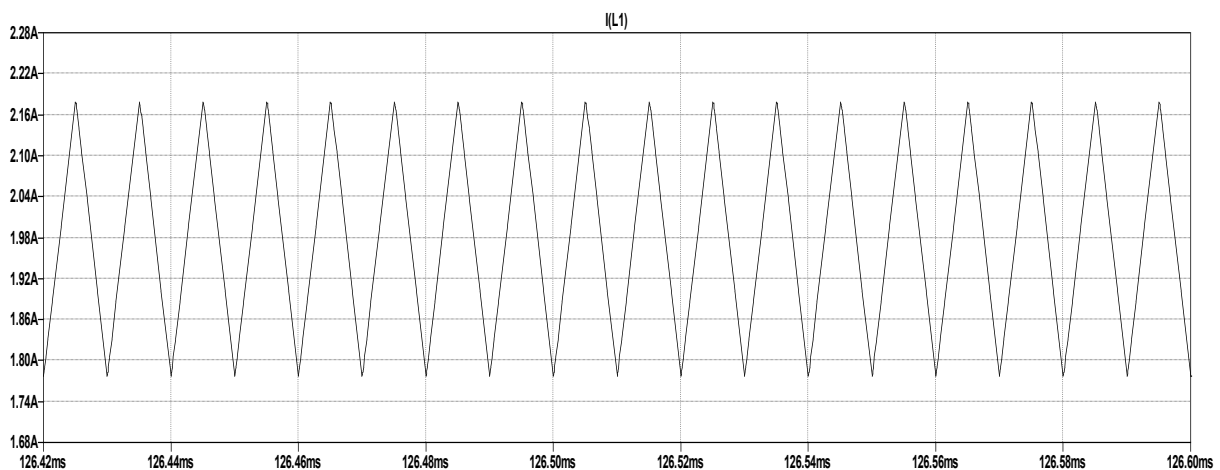


Figure 2.8 Steady-State Inductor Current Waveform Obtained from LTspice

The LTspice simulation provides a visual confirmation of the converter operation and demonstrates the expected switching behaviour of the practical circuit. Since LTspice was primarily used for circuit-level validation, the steady-state

performance parameters such as average values, RMS values and ripple magnitudes were not extracted numerically in this work. These quantities were evaluated more accurately using MATLAB/Simulink and the accompanying MATLAB analysis scripts presented in the previous section.

The close agreement between the LTspice waveforms and the MATLAB/Simulink results confirms the correctness of the converter design and provides additional confidence in the analytical calculations and selected component values.

The analytical calculations, MATLAB/Simulink verification and LTspice circuit-level validation collectively confirm that the selected converter parameters satisfy the design requirements. Having verified the open-loop operation of the converter, the next step is to develop its mathematical model. This model forms the basis for the design of the closed-loop PI controller presented in the following chapter.

3. Mathematical Modelling

3.1 Introduction and Modelling Assumptions

The design of a closed-loop controller requires a mathematical model that accurately describes the dynamic behaviour of the buck converter. Since the converter operates by rapidly switching semiconductor devices, its behaviour is inherently nonlinear and time-varying. Direct controller design using the switching model is therefore difficult. To facilitate controller design and stability analysis, the converter is represented by a linearized small-signal model about its steady-state operating point.

The small-signal model is obtained using the state-space averaging technique, where the converter dynamics during the ON and OFF switching intervals are averaged over one switching period. The averaged model is then linearized by introducing small perturbations around the steady-state operating point. This results in a linear time-invariant model that accurately represents the converter dynamics for small variations around the nominal operating condition.

The mathematical model developed in this chapter forms the basis for the design of the PI controller presented in the subsequent chapter.

The following assumptions are made during the modelling process:

- The converter operates in Continuous Conduction Mode (CCM).
- The MOSFET switch and freewheeling diode are considered ideal.
- The inductor winding resistance is neglected.
- The equivalent series resistance (ESR) of the output capacitor is neglected.
- The switching frequency remains constant throughout the analysis.
- Only small perturbations about the steady-state operating point are considered, allowing higher-order perturbation terms to be neglected.

3.2 Small-Signal Modelling of the Buck Converter

The state-space averaging method is employed to obtain the mathematical model of the buck converter. Since the inductor and capacitor are the energy storage elements of the converter, the inductor current and capacitor voltage are selected as the state variables because they uniquely define the state of the system.

Accordingly, the state vector is defined as

$$x = \begin{bmatrix} i_L \\ v_C \end{bmatrix}$$

where i_L represents the inductor current and v_C represents the capacitor voltage.

The operation of the buck converter consists of two switching intervals within one switching period.

Switch ON State ($0 < t < DT$)

When the MOSFET is turned ON, the input source supplies energy to both the load and the inductor, while the freewheeling diode remains reverse biased. Applying Kirchhoff's Voltage Law (KVL) and Kirchhoff's Current Law (KCL) yields

$$\begin{aligned} v_L &= V_g - v_C \\ i_C &= i_L - \frac{v_C}{R} \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{di_L}{dt} &= \frac{V_g - v_C}{L} \\ \frac{dv_C}{dt} &= \frac{1}{C} \left(i_L - \frac{v_C}{R} \right) \end{aligned}$$

Switch OFF State ($DT < t < T$)

When the MOSFET is turned OFF, the input source is disconnected from the converter and the inductor releases its stored energy through the freewheeling diode to the load. During this interval,

$$\begin{aligned} v_L &= -v_C \\ i_C &= i_L - \frac{v_C}{R} \end{aligned}$$

which gives

$$\begin{aligned}\frac{di_L}{dt} &= -\frac{v_C}{L} \\ \frac{dv_C}{dt} &= \frac{1}{C}\left(i_L - \frac{v_C}{R}\right)\end{aligned}$$

The converter behaviour over one switching period is obtained by averaging the ON-state and OFF-state equations using the duty ratio d . The averaged state equations are therefore expressed as

$$\begin{aligned}\frac{di_L}{dt} &= \frac{1}{L}(dV_g - v_C) \\ \frac{dv_C}{dt} &= \frac{1}{C}\left(i_L - \frac{v_C}{R}\right)\end{aligned}$$

To obtain the small-signal model, each converter variable is expressed as the sum of its steady-state value and a small perturbation,

$$\begin{aligned}v_g &= V_g + \tilde{v}_g \\ i_L &= I_L + \tilde{i}_L \\ v_C &= V_C + \tilde{v}_C \\ d &= D + \tilde{d}\end{aligned}$$

where the uppercase quantities denote the steady-state operating values and the variables with a tilde represent small perturbations about the operating point.

Substituting these expressions into the averaged state equations and neglecting second-order perturbation terms yields the linearized small-signal equations

$$\begin{aligned}\frac{d\tilde{i}_L}{dt} &= \frac{1}{L}(D\tilde{v}_g + V_g\tilde{d} - \tilde{v}_C) \\ \frac{d\tilde{v}_C}{dt} &= \frac{1}{C}\left(\tilde{i}_L - \frac{\tilde{v}_C}{R}\right)\end{aligned}$$

The above equations may be represented in state-space form as

$$\dot{x} = Ax + B_c u_c + B_d u_d$$

where the control input is the duty-ratio perturbation $\tilde{d}$ and the disturbance input is the input-voltage perturbation $\tilde{v}_g$.

The state-space model obtained above is used in the following section to derive the transfer functions required for controller design.

3.3 Small-Signal Transfer Function

The linearized state-space model obtained in the previous section is transformed into the Laplace domain to derive the small-signal transfer functions of the buck converter. These transfer functions describe the dynamic relationship between the output voltage and the various perturbation inputs acting on the converter.

The linearized state-space equations are

$$\dot{x} = Ax + B_c \tilde{d} + B_d \tilde{V}_g$$

Taking the Laplace transform,

$$sX(s) = AX(s) + B_c \tilde{D}(s) + B_d \tilde{V}_g(s)$$

Rearranging,

$$(sI - A)X(s) = B_c \tilde{D}(s) + B_d \tilde{V}_g(s)$$

Hence,

$$X(s) = (sI - A)^{-1} [B_c \tilde{D}(s) + B_d \tilde{V}_g(s)]$$

The output of the converter is the capacitor voltage. Therefore,

$$\tilde{V}_o(s) = CX(s)$$

Substituting the state vector into the above equation gives

$$\tilde{V}_o(s) = C(sI - A)^{-1} B_c \tilde{D}(s) + C(sI - A)^{-1} B_d \tilde{V}_g(s)$$

Comparing with the standard transfer-function form,

$$\tilde{V}_o(s) = G_{vd}(s) \tilde{D}(s) + G_{vg}(s) \tilde{V}_g(s)$$

where

$$G_{vd}(s) = C(sI - A)^{-1} B_c$$

and

$$G_{vg}(s) = C(sI - A)^{-1} B_d$$

After simplification, the control-to-output transfer function is obtained as

$$G_{vd}(s) = \frac{\tilde{V}_o(s)}{\tilde{D}(s)} = \frac{\frac{V_g}{LC}}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

Similarly, the input-to-output transfer function becomes

$$G_{vg}(s) = \frac{\tilde{V}_o(s)}{\tilde{V}_g(s)} = \frac{\frac{D}{LC}}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}$$

In general, the complete small-signal output voltage of a buck converter can be represented as

$$\boxed{\tilde{V}_o(s) = G_{vd}(s)\tilde{D}(s) + G_{vg}(s)\tilde{V}_g(s) + Z_o(s)\tilde{I}_{load}(s)}$$

where

- $G_{vd}(s)$ represents the effect of duty-ratio perturbations,
- $G_{vg}(s)$ represents the effect of input-voltage perturbations, and
- $Z_o(s)$ represents the output impedance relating load-current disturbances to the output voltage.

Thus, the output voltage of the converter is influenced by three independent inputs: the duty ratio, the input supply voltage and the load current.

In this project, the converter output voltage is regulated by controlling the PWM duty ratio. Therefore, the control-to-output transfer function is selected as the plant for controller design. The remaining two terms represent external disturbances and are used to explain the converter behaviour during the line regulation and load regulation tests presented in Chapter 6.

Substituting the converter parameters obtained in Chapter 2,

$$V_g = 24 \text{ V}, L = 150 \mu\text{H}, C = 10 \mu\text{F}, R = 6 \Omega$$

the numerical plant transfer function becomes

$$G_{vd}(s) = \frac{1.6 \times 10^{10}}{s^2 + 1.6667 \times 10^4 s + 6.667 \times 10^8}$$

This transfer function is imported into MATLAB and forms the mathematical plant used for the PI controller design presented in the following chapter.

4. PI Controller Design

4.1 Controller Design Specifications

The control-to-output transfer function derived in Chapter 3 represents the mathematical model of the buck converter and serves as the plant for controller design. The objective of the controller is to regulate the output voltage while providing fast transient response, high stability, and accurate steady-state performance under varying operating conditions.

To achieve these objectives, the controller is designed using frequency-domain techniques. Frequency-domain design enables the stability and dynamic performance of the closed-loop system to be evaluated through parameters such as gain margin, phase margin, bandwidth and crossover frequency. The controller is therefore designed to satisfy the following performance specifications.

- **Settling Time (T_s):** The closed-loop system is required to achieve a settling time of 0.5 ms to ensure rapid regulation of the output voltage following a disturbance or reference change.
- **Steady-State Error (e_{ss}):** The controller should provide zero steady-state error for a step reference input to ensure accurate output voltage regulation.
- **Relative Stability:** The closed-loop system should possess adequate stability margins to obtain a well-damped transient response with minimal overshoot.
- **Bandwidth:** The controller bandwidth should be selected to provide a fast dynamic response while remaining sufficiently lower than the converter switching frequency.
- **High-Frequency Attenuation:** Adequate attenuation at higher frequencies is desirable to reduce the influence of switching harmonics and measurement noise on the control system.

These design specifications serve as the basis for analysing the uncompensated buck converter and designing the PI controller in the following sections.

4.2 Frequency-Domain Analysis of the Uncompensated Plant

The control-to-output transfer function derived in Chapter 3 was imported into MATLAB for frequency-domain analysis. The uncompensated plant was analysed using the Bode plot to evaluate its gain characteristics, phase response and stability before designing the controller.

The numerical transfer function of the buck converter is

$$G_{vd}(s) = \frac{1.6 \times 10^{10}}{s^2 + 1.667 \times 10^4 s + 6.667 \times 10^8}$$

For better interpretation of the system dynamics, the transfer function may also be expressed in the standard second-order form

$$G_{vd}(s) = \frac{K}{\left(\frac{s}{\omega_n}\right)^2 + \frac{1}{Q}\left(\frac{s}{\omega_n}\right) + 1}$$

$$G_{vd}(s) = \frac{24}{\left(\frac{s}{25820}\right)^2 + \frac{1}{1.55}\left(\frac{s}{25820}\right) + 1}$$

where K is the DC gain, ω_n is the natural frequency and Q is the quality factor of the system.

The open-loop Bode plot of the uncompensated plant is shown in Figure 4.1.

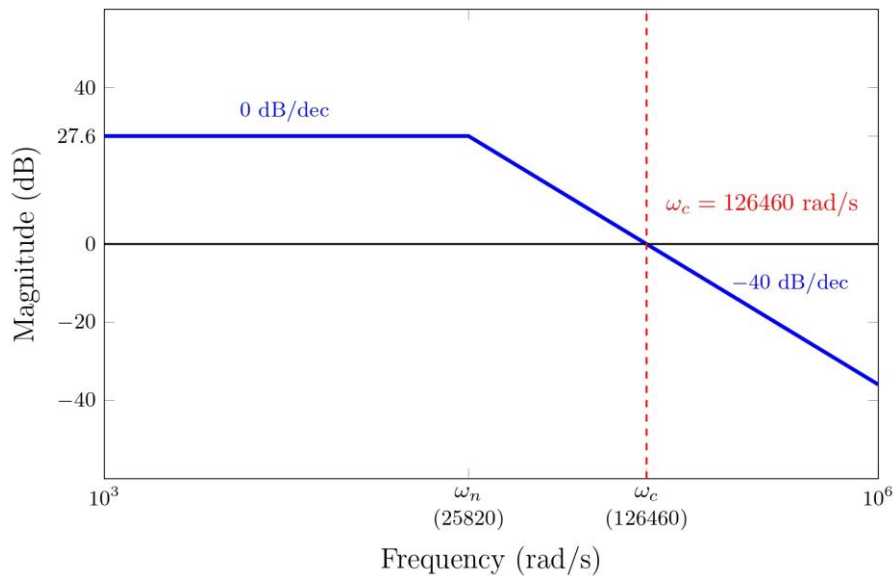


Figure 4.1 Asymptotic Open-Loop Bode Plot of the Uncompensated Buck Converter

From the Bode plot, the converter exhibits a low-frequency gain of approximately 27.6 dB. The magnitude response begins to decrease with a slope of approximately -40 dB/decade beyond the natural frequency of the converter, indicating the presence of two dominant poles.

The uncompensated plant has a gain crossover frequency of approximately 126.46 krad/s, which is considerably higher than the bandwidth required to satisfy the desired transient performance. Although the plant is stable, its frequency response does not directly satisfy the controller design specifications established in the previous section.

Therefore, compensation is required to modify the loop characteristics and obtain the desired crossover frequency while maintaining adequate stability margins. A PI controller is selected for this purpose, and its design is presented in the following section.

4.3 PI Controller Design

The frequency-domain analysis presented in the previous section shows that the uncompensated buck converter does not satisfy the desired controller specifications. Therefore, a suitable compensator is required to modify the loop characteristics and obtain the required transient response while maintaining adequate stability margins.

A Proportional-Integral (PI) controller is selected for this work due to its simplicity and its ability to eliminate steady-state error for a step reference input. The proportional term improves the transient response, whereas the integral term increases the low-frequency loop gain, thereby ensuring zero steady-state error.

The transfer function of the PI controller is

$$C(s) = K_p + \frac{K_i}{s}$$

which may also be written as

$$C(s) = K_i \frac{\left(\frac{s}{K_i/K_p} + 1\right)}{s}$$

where K_p is the proportional gain and K_i is the integral gain.

From the controller specifications presented in Section 4.1, the desired settling time is

$$T_s = 0.5 \text{ ms}$$

Using the approximation

$$\omega_c \approx \frac{4}{T_s}$$

the desired gain crossover frequency is obtained as

$$\omega_c \approx \frac{4}{0.5 \times 10^{-3}} = 8000 \text{ rad/s}$$

The uncompensated plant, however, has a gain crossover frequency of approximately 126.46 krad/s, which is significantly higher than the desired value. Therefore, the controller must reduce the crossover frequency while preserving adequate phase margin.

Since the PI controller introduces a pole at the origin and one finite zero, the location of the zero must be selected carefully. Placing the zero below the desired crossover frequency would flatten the magnitude response before crossover, resulting in an undesirable loop shape for the present design. Therefore, the controller zero is selected beyond the desired crossover frequency.

For this project, the PI controller zero is placed at

$$\omega_z = 20000 \text{ rad/s}$$

Accordingly,

$$\frac{K_i}{K_p} = 20000$$

The required controller gain is obtained from the magnitude plot of the uncompensated plant. At the desired crossover frequency of 8000 rad/s, the magnitude of the plant is approximately -50.46 dB. Therefore, the PI controller must provide a gain of 50.46 dB at this frequency so that the compensated open-loop gain becomes 0 dB, thereby establishing the desired gain crossover frequency.

Converting the gain into linear form gives

$$K_i = 333.5$$

Using the selected zero location,

$$K_p = \frac{K_i}{20000} = 0.016675$$

Hence, the final PI controller is obtained as

$$C(s) = 333.5 \frac{\left(\frac{s}{20000} + 1\right)}{s}$$

The asymptotic Bode plot of the compensated system is shown in Figure 4.2, illustrating the effect of the PI controller on the loop gain and crossover frequency.

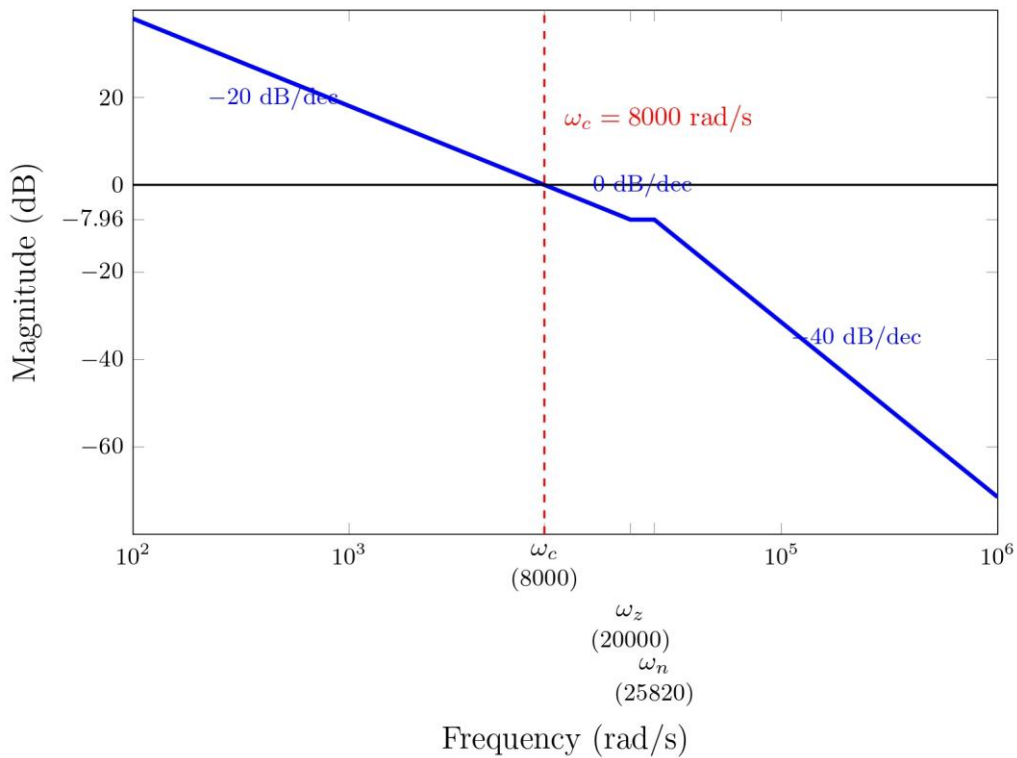


Figure 4.2 Asymptotic Bode Plot of the Compensated System

The designed controller was subsequently implemented and verified in MATLAB. The compensated open-loop Bode plot is shown in Figure 4.3, while the corresponding closed-loop step response is presented in Figure 4.4.

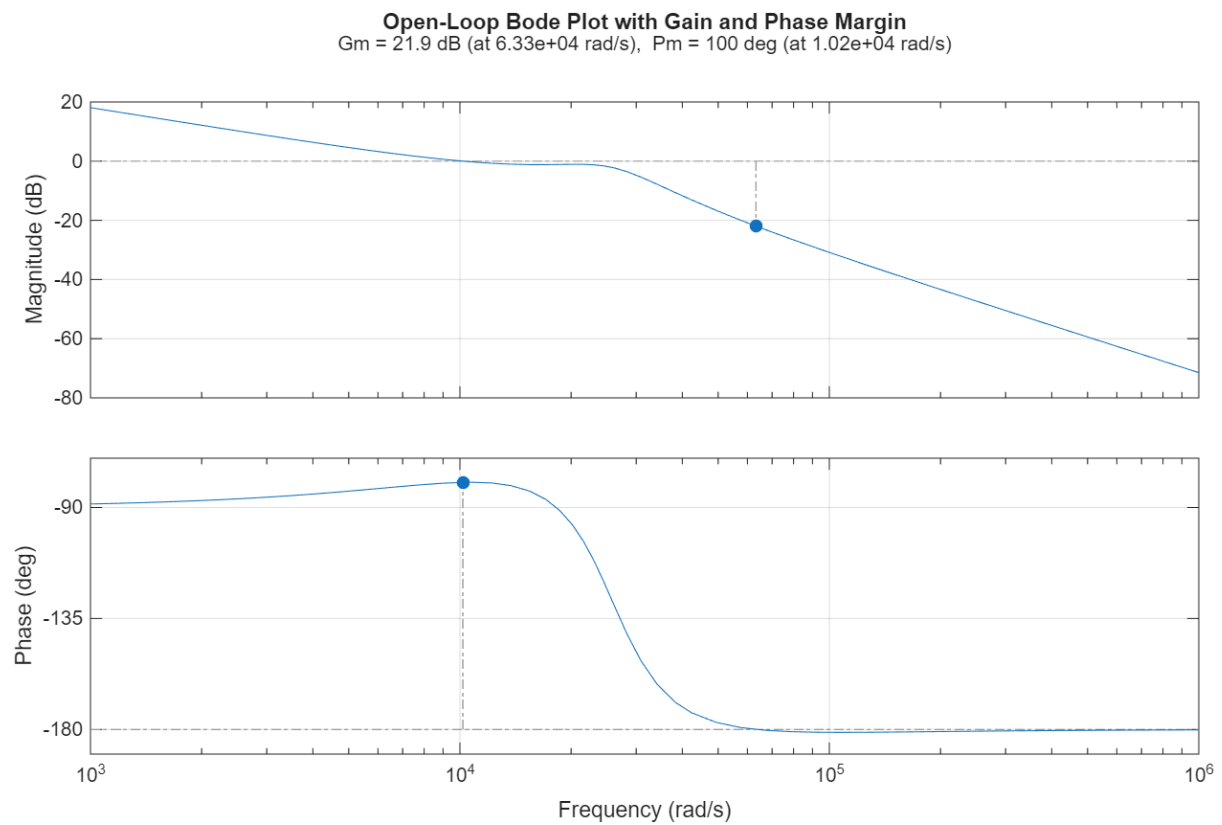


Figure 4.3 MATLAB Bode Plot of the Compensated System

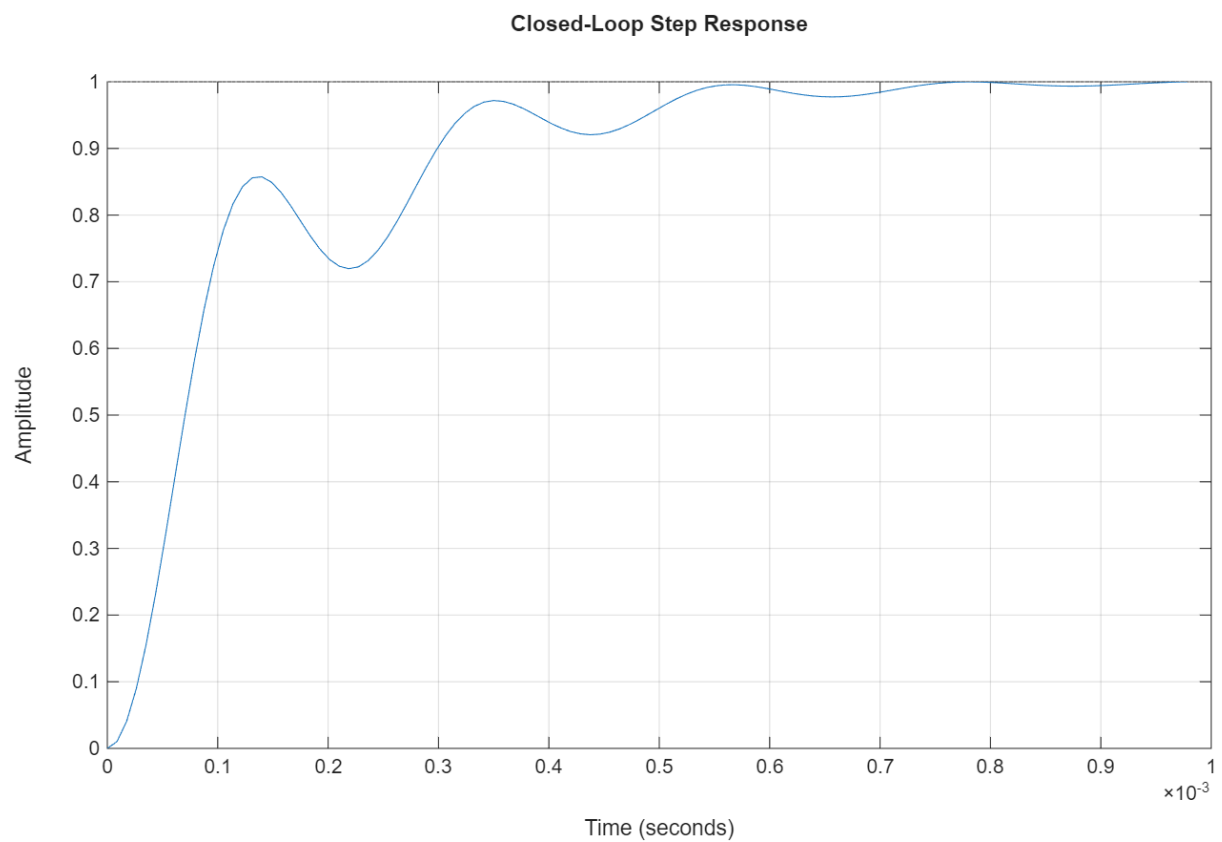


Figure 4.4 Closed-Loop Step Response of the Compensated System

The corresponding stability margins and transient response characteristics obtained from MATLAB are summarized in Table 4.1. The results confirm that the designed controller satisfies the specified stability and transient performance requirements and is therefore selected for the closed-loop implementation presented in the next chapter.

Table 4.1 Performance Parameters of the Designed PI Controller

Parameter	Value
Gain Margin (dB)	21.94
Phase Margin (°)	100.21
Gain Crossover Frequency (rad/s)	10183.68
Phase Crossover Frequency (rad/s)	63256.39
Rise Time (ms)	0.271
Settling Time (ms)	0.682
Peak Time (ms)	0.998
Overshoot (%)	0.027

The MATLAB implementation used for frequency-domain analysis, PI controller design, stability margin calculation and closed-loop step response evaluation is provided in Appendix C. The script generates the Bode plots, computes the gain margin, phase margin, crossover frequencies and transient response characteristics summarized in Table 4.1, thereby enabling reproducibility of the controller design presented in this chapter.

The compensated system satisfies the desired stability and transient response specifications obtained during the controller design stage. Having verified the controller performance through frequency-domain analysis and time-domain simulation, the next step is to implement the designed PI controller in the complete closed-loop buck converter model developed in MATLAB/Simulink.

5. Closed-Loop Implementation

5.1 Closed-Loop Simulink Model

The complete closed-loop buck converter was implemented in MATLAB/Simulink using the power stage developed in Chapter 2 together with the PI controller designed in Chapter 4. The objective of the closed-loop implementation is to automatically regulate the output voltage by continuously adjusting the converter duty ratio based on the measured output voltage.

The complete closed-loop Simulink model is shown in Figure 5.1.

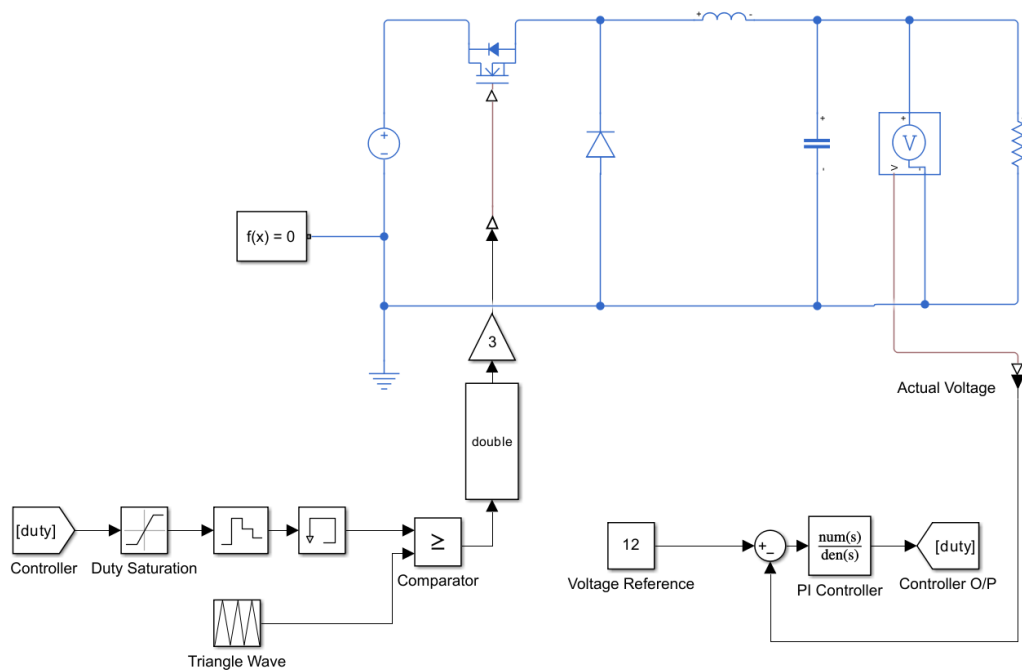


Figure 5.1 Complete Closed-Loop Buck Converter Model Implemented in MATLAB/Simulink

The closed-loop system operates by continuously comparing the measured output voltage with the desired reference voltage. The difference between these two signals produces the error signal, which is applied to the PI controller. Based on the magnitude of this error, the controller generates the required duty ratio for the buck converter.

Since the control-to-output transfer function derived in Chapter 3 uses the duty ratio as the control input, the output of the PI controller directly represents the required duty cycle. The controller output is subsequently processed to generate the PWM switching signal, which drives the MOSFET of the buck converter. The

resulting output voltage is measured using a voltage sensor and fed back to the summing junction, thereby completing the closed feedback loop.

Through this continuous feedback process, the controller automatically adjusts the duty ratio to maintain the desired output voltage despite changes in the operating conditions. The detailed implementation of the controller and PWM generation is described in the following section.

5.2 Controller Implementation

The PI controller developed in Chapter 4 was implemented directly in the closed-loop Simulink model. Since the control-to-output transfer function of the buck converter uses the duty ratio as its input, the output of the PI controller directly represents the required duty cycle needed to regulate the output voltage.

The duty ratio generated by the controller is passed through a Saturation block before PWM generation. This limits the controller output between 0 and 1, ensuring that the duty ratio remains within the physically realizable operating range of the buck converter. Without this limitation, the controller may command duty ratios beyond the allowable range during transient conditions, resulting in unrealistic converter operation.

The limited duty ratio is then compared with a high-frequency triangular carrier waveform to generate the PWM switching signal. Whenever the controller duty ratio is greater than the instantaneous value of the triangular carrier, the comparator output becomes high and the MOSFET is switched ON. Otherwise, the MOSFET remains OFF. This process produces a PWM signal whose duty cycle is continuously adjusted by the controller according to the output voltage error.

A Zero-Order Hold (ZOH) block is introduced between the controller and the PWM generation stage. The ZOH updates the controller output once every switching period, thereby representing the digital implementation of the controller. Since the converter switching frequency is 100 kHz, the sampling period of the ZOH is selected as

$$T_s = \frac{1}{100 \times 10^3} = 10 \mu s$$

A Memory block is also incorporated in the control loop to eliminate the algebraic loop created by the direct feedback path in Simulink. The Memory block introduces a one-sample delay, allowing the controller output to be computed

using the previous simulation step while preserving the intended closed-loop behaviour.

Finally, the converter output voltage is measured using a voltage sensor and fed back to the summing junction, where it is continuously compared with the reference voltage. This feedback mechanism enables the PI controller to automatically modify the duty ratio whenever disturbances occur, thereby maintaining the desired output voltage.

5.3 Implementation Verification

Before evaluating the controller under different operating conditions, the complete closed-loop implementation was verified to ensure the correct operation of the control algorithm and PWM generation. The objective of this verification was not to evaluate the controller performance in detail, but to confirm that the implemented model operated correctly over the practical operating range of the converter.

The closed-loop model was tested for different reference voltages within the achievable output voltage range of the converter. The controller successfully regulated the output voltage from approximately 0.5 V to 23.96 V, demonstrating stable operation over almost the entire operating range corresponding to the 24 V input supply.

These preliminary simulations confirmed the correct implementation of the PI controller, PWM generation, feedback path and duty ratio control. Consequently, the developed closed-loop model was considered suitable for detailed performance evaluation.

The performance of the controller under various operating conditions, including converter performance, reference tracking, line regulation and load regulation, is presented in the following chapter.

6. Performance Evaluation and Validation

6.1 Steady-State Converter Performance

The first validation test was carried out to evaluate the steady-state performance of the closed-loop buck converter under its rated operating conditions. The converter was operated with an input voltage of 24 V, an output voltage reference of 12 V, and a rated load resistance of 6 Ω . The objective of this test was to verify that the designed converter satisfies the required output voltage, current, and ripple specifications during steady-state operation.

The output voltage, output current and inductor current were analysed after the converter reached steady state. The average values, ripple magnitudes and RMS values obtained from the simulation are summarized in Table 6.1.

Table 6.1 Steady-State Performance of the Closed-Loop Buck Converter

Parameter	Value
Average Output Voltage	11.9991 V
Average Output Current	1.9998 A
Average Inductor Current	2.0056 A
Output Voltage Ripple (p-p)	45.34 mV
Inductor Current Ripple (p-p)	0.4022 A
Output Voltage RMS	11.9991 V
Output Current RMS	1.9998 A
Inductor Current RMS	2.0103 A

In addition to the steady-state characteristics, the transient response of the closed-loop converter was evaluated at the rated output voltage of 12 V. The measured performance parameters are summarized in Table 6.2. The converter achieves a rise time of 0.272 ms and a settling time of 0.702 ms, both of which satisfy the desired transient response requirements established during the controller design stage. The overshoot is limited to 0.534%, while the steady-state error is only 0.00071 V (0.00593%), confirming excellent voltage regulation at the nominal operating condition.

Table 6.2 Closed-Loop Performance at the Rated Operating Condition

Parameter	Value
Reference Voltage	12.000 V
Steady-State Output	11.999 V
Rise Time	0.272 ms
Settling Time	0.702 ms
Overshoot	0.534 %
Steady-State Error	0.00071 V
Steady-State Error (%)	0.00593 %

The steady-state output voltage and inductor current waveforms are shown in Figure 6.1 and Figure 6.2, respectively.

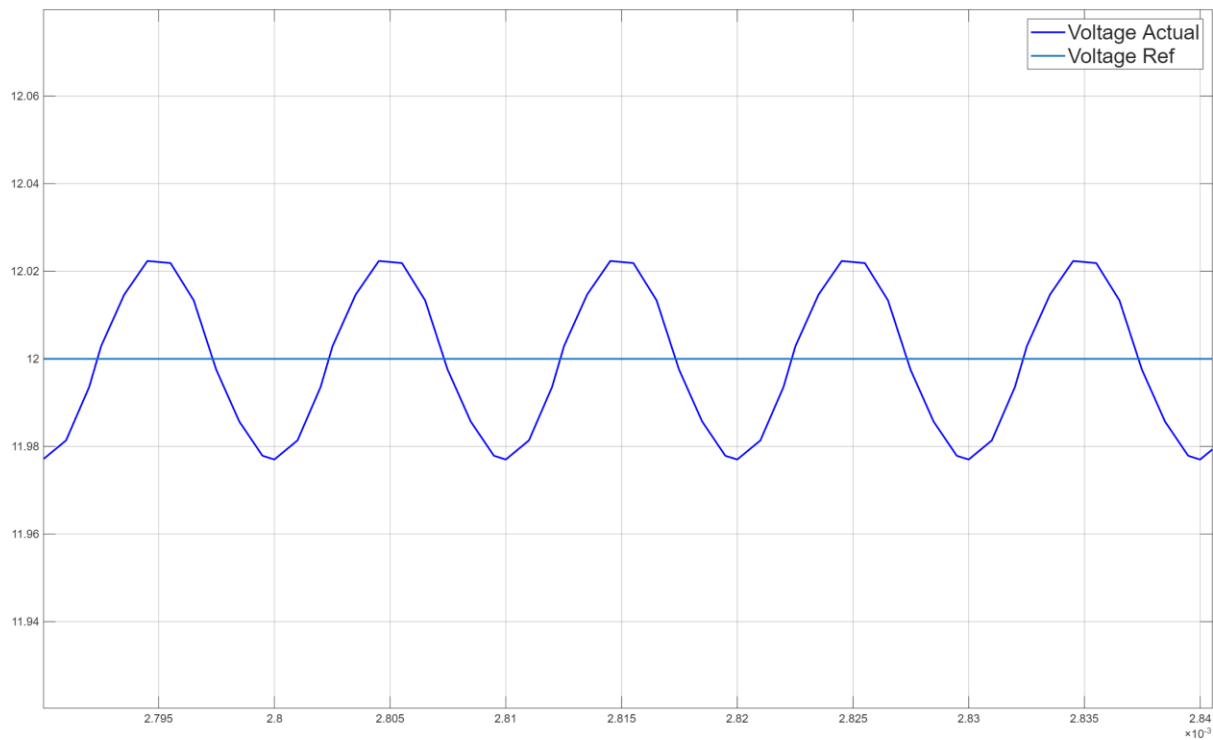


Figure 6.1 Steady-State Output Voltage of the Closed-Loop Buck Converter

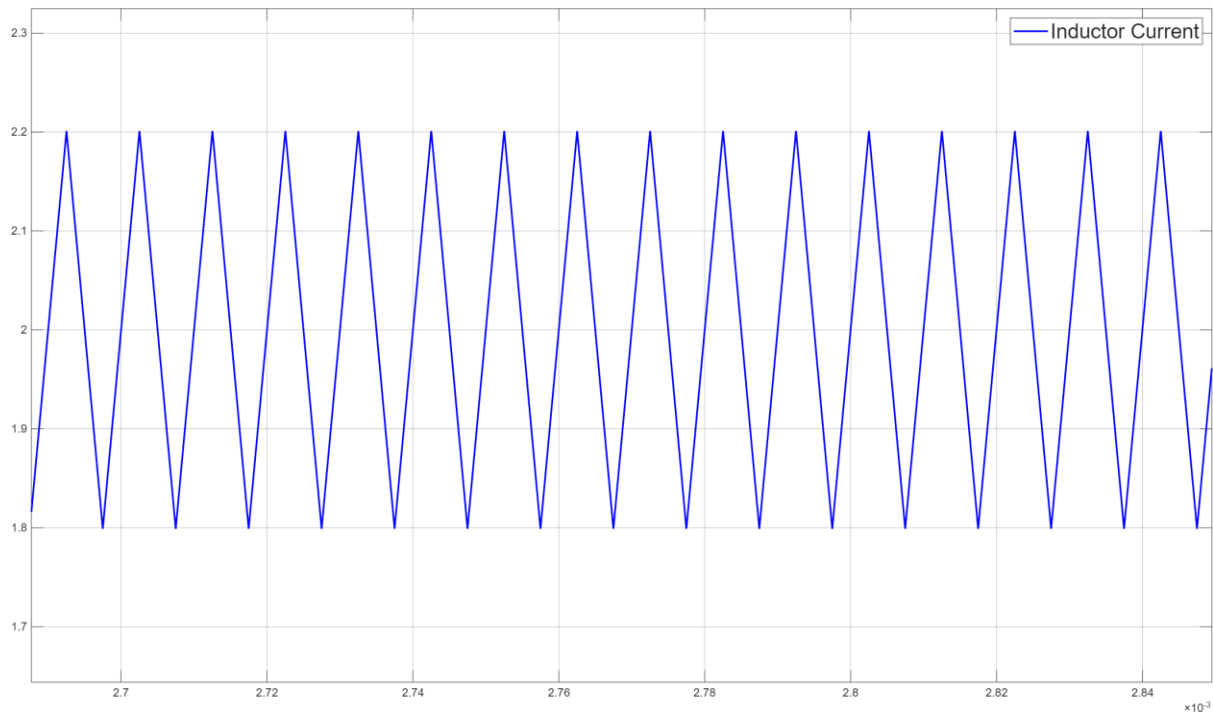


Figure 6.2 Steady-State Inductor Current of the Closed-Loop Buck Converter

From Table 6.1, the average output voltage is 11.9991 V, which is very close to the desired reference voltage of 12 V, demonstrating excellent voltage regulation. The average output current of 1.9998 A also closely matches the rated load current, confirming correct converter operation under nominal conditions.

The measured output voltage ripple is approximately 45.34 mV (peak-to-peak), which satisfies the design target of less than 60 mV. Similarly, the measured inductor current ripple is 0.4022 A (peak-to-peak), which closely agrees with the design value of 0.4 A obtained during the converter design stage. These results verify the correctness of the selected inductor and capacitor values.

The RMS values of the output voltage, output current and inductor current further confirm stable converter operation with negligible deviation from the desired operating point.

The MATLAB script used to calculate the average values, ripple magnitudes and RMS values presented in Table 6.1 is provided in Appendix D.

The steady-state performance obtained in this section confirms that the closed-loop buck converter satisfies the specified design requirements under nominal operating conditions. The subsequent sections evaluate the controller performance under different reference voltages, input voltage disturbances and load variations.

6.2 Static Reference Tracking

The static reference tracking capability of the designed PI controller was evaluated by applying different constant reference voltages to the closed-loop buck converter. The objective of this test was to verify that the controller accurately regulates the output voltage over a wide operating range while maintaining satisfactory transient and steady-state performance.

Four reference voltages of 3.3 V, 5 V, 12 V and 22 V were selected to represent low, medium, rated and high output voltage conditions. For each reference value, the steady-state output voltage, rise time, settling time, overshoot and steady-state error were obtained from the simulation. The results are summarized in Table 6.3.

Table 6.3 Static Reference Tracking Performance of the Closed-Loop Buck Converter

Reference (V)	Steady- State Output (V)	Rise Time (ms)	Settling Time (ms)	Overshoot (%)	Steady- State Error (V)	Error (%)
3.3	3.295	0.273	0.720	0.795	0.00463	0.14017
5.0	4.996	0.273	0.710	0.626	0.00437	0.08741
12.0	11.999	0.272	0.702	0.534	0.00071	0.00593
22.0	22.002	0.272	0.700	0.412	0.00216	0.00983

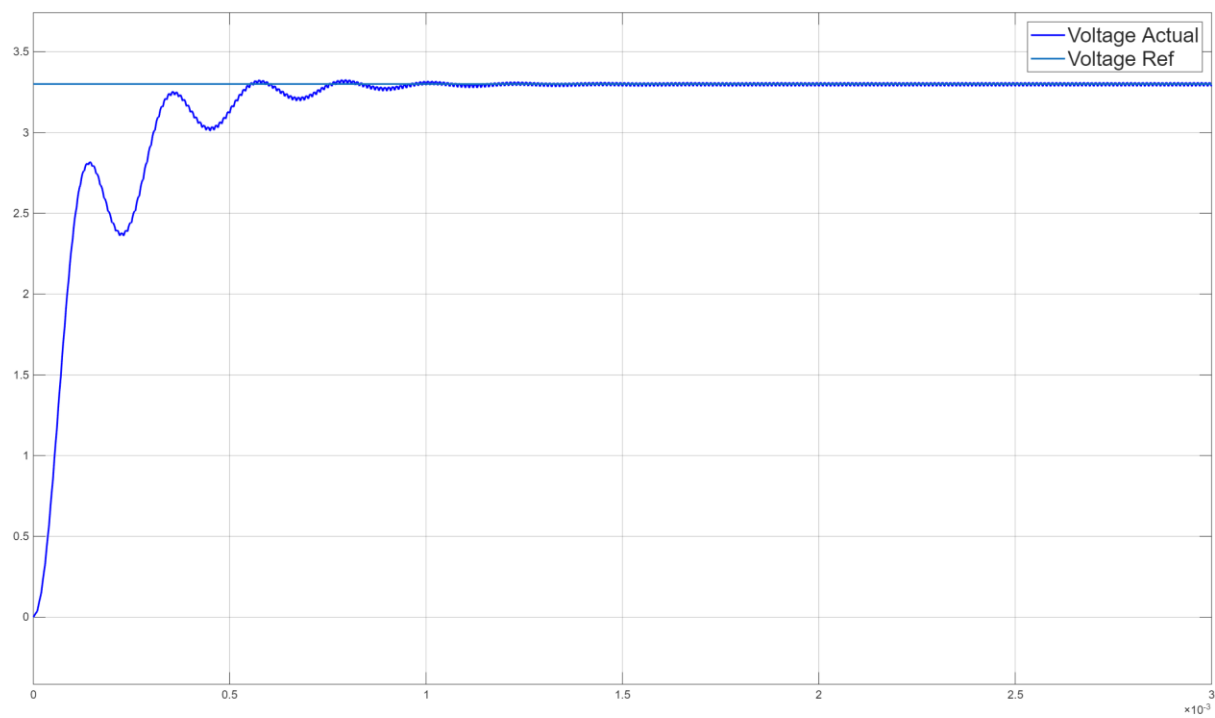


Figure 6.3 Static Reference Tracking Response for a 3.3 V Reference

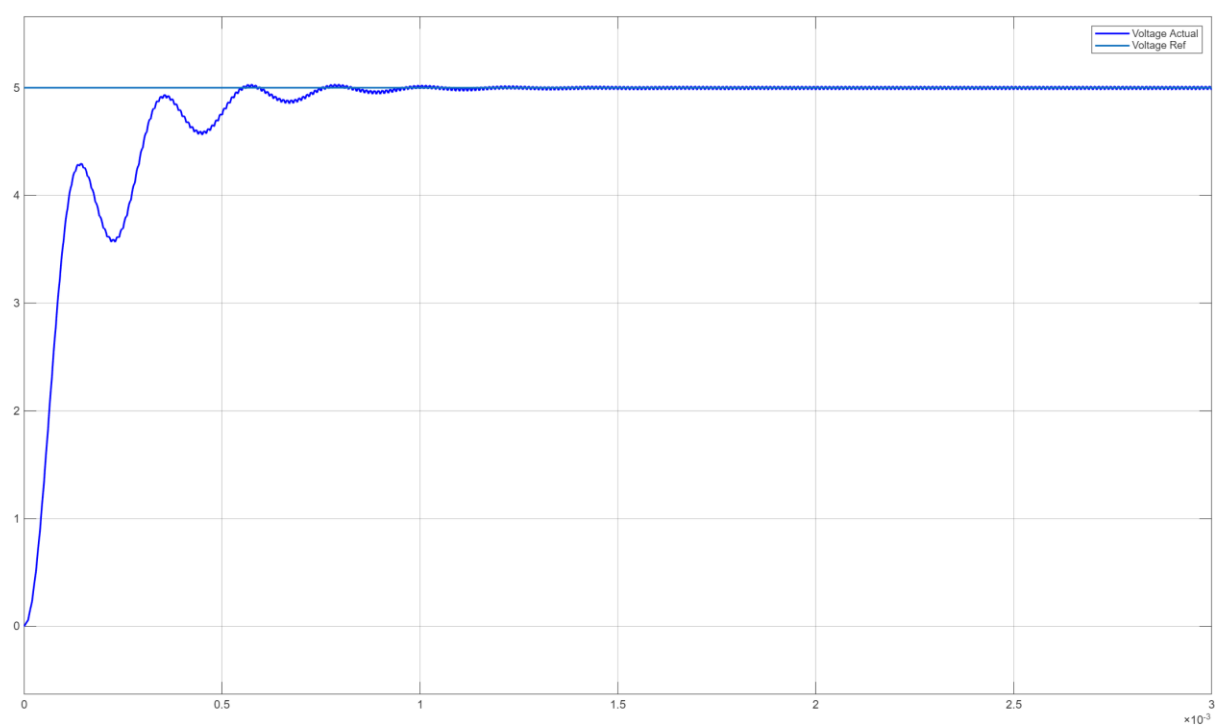


Figure 6.4 Static Reference Tracking Response for a 5 V Reference

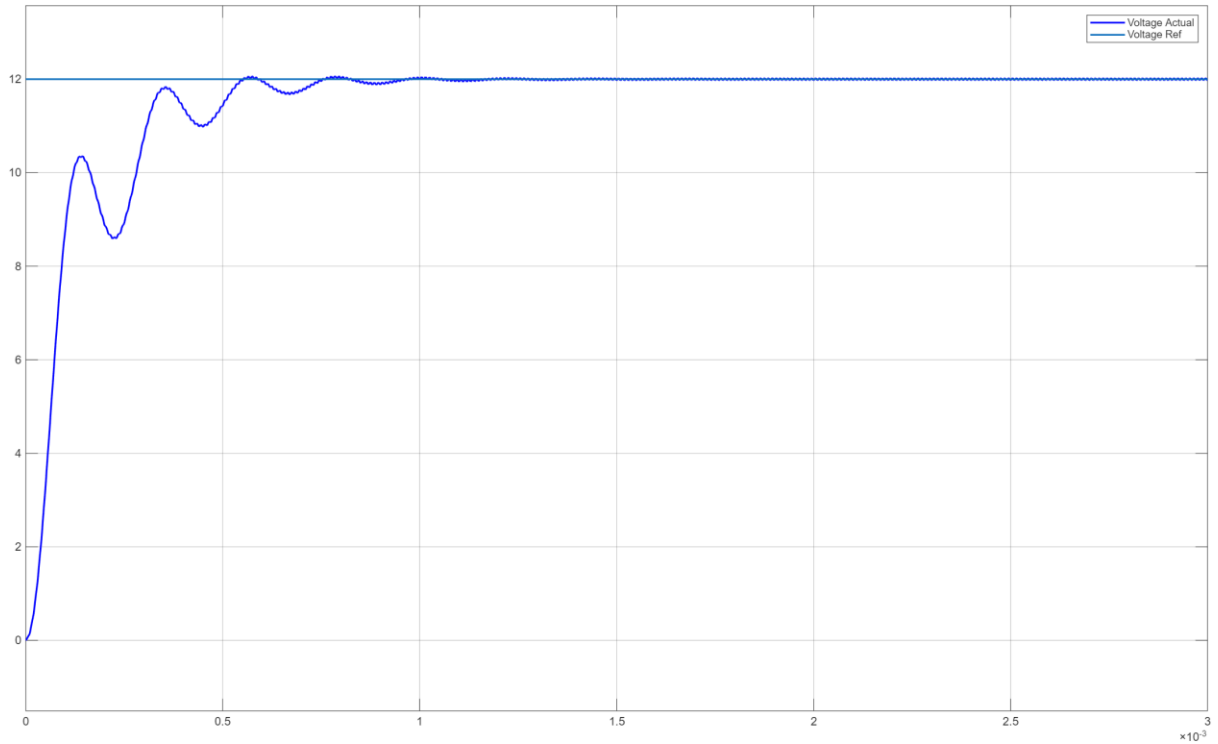


Figure 6.5 Static Reference Tracking Response for a 12 V Reference

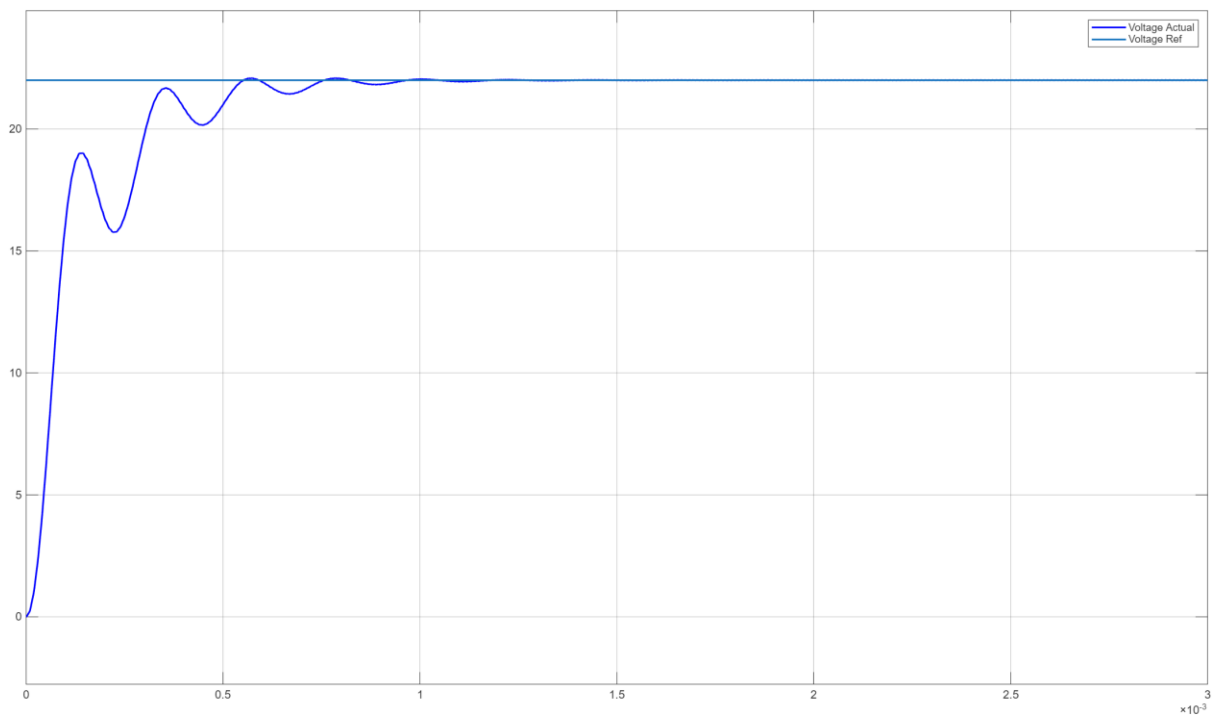


Figure 6.6 Static Reference Tracking Response for a 22 V Reference

From Table 6.3, it is observed that the designed PI controller accurately regulates the output voltage over a wide operating range. In all four cases, the measured steady-state output closely follows the applied reference voltage with very small steady-state error.

The rise time and settling time remain nearly constant for all reference voltages, indicating that the controller dynamics are largely independent of the operating point. Similarly, the overshoot remains below 1% in every case, demonstrating a well-damped transient response.

These results confirm that the designed controller provides accurate reference tracking while maintaining fast transient response and excellent steady-state performance over almost the entire operating range of the converter.

The MATLAB script used to evaluate the static reference tracking performance is provided in Appendix E. The same program was used for all test cases, with only the reference voltage value being modified.

6.3 Dynamic Reference Tracking

The dynamic reference tracking performance of the closed-loop buck converter was evaluated by applying successive step changes to the reference voltage while maintaining a constant input voltage and load resistance. Unlike the static reference tracking test presented in the previous section, this test examines the controller's ability to respond to continuously changing operating conditions without requiring the system to be restarted for each reference value.

The reference voltage was initially set to 3.3 V, then increased to 22 V, and finally reduced to 5 V. The corresponding output voltage response is shown in Figure 6.7.

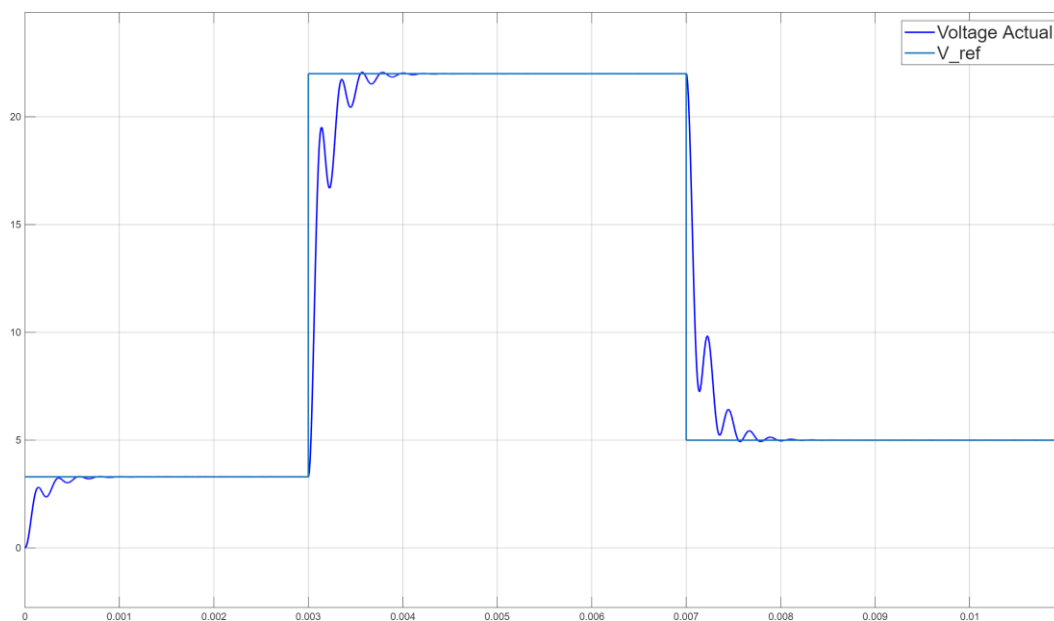


Figure 6.7 Dynamic Reference Tracking Response of the Closed-Loop Buck Converter

It can be observed that the output voltage closely follows each reference step with a fast transient response and negligible steady-state error. During both the increase and decrease in reference voltage, the controller rapidly adjusts the duty ratio to regulate the converter output to the newly commanded value.

Small transient oscillations are observed immediately after each reference step due to the energy exchange between the inductor and capacitor of the output filter. However, these oscillations are well damped and decay rapidly, indicating that the designed PI controller provides adequate damping while maintaining a fast response.

The results demonstrate that the closed-loop buck converter is capable of accurately tracking rapidly changing reference voltages over a wide operating range. This confirms the effectiveness of the controller in providing stable and robust reference tracking while preserving satisfactory transient and steady-state performance.

6.4 Line Regulation

The line regulation capability of the closed-loop buck converter was evaluated by varying the input supply voltage while maintaining a constant reference voltage of 12 V and a constant load resistance of 6 Ω . The objective of this test was to verify the ability of the designed PI controller to reject input voltage disturbances and maintain a regulated output voltage.

To implement the test in MATLAB/Simulink, the fixed DC input source of the converter was replaced with a Controlled Voltage Source. The input of the controlled source was driven by a Step block to produce successive changes in the supply voltage during the simulation. The supply voltage was varied from 24 V to 20 V, then increased to 28 V, and finally reduced to 18 V, while the output voltage reference remained fixed at 12 V throughout the test.

The resulting output voltage response is shown in Figure 6.8, while the corresponding duty ratio generated by the PI controller is shown in Figure 6.9.

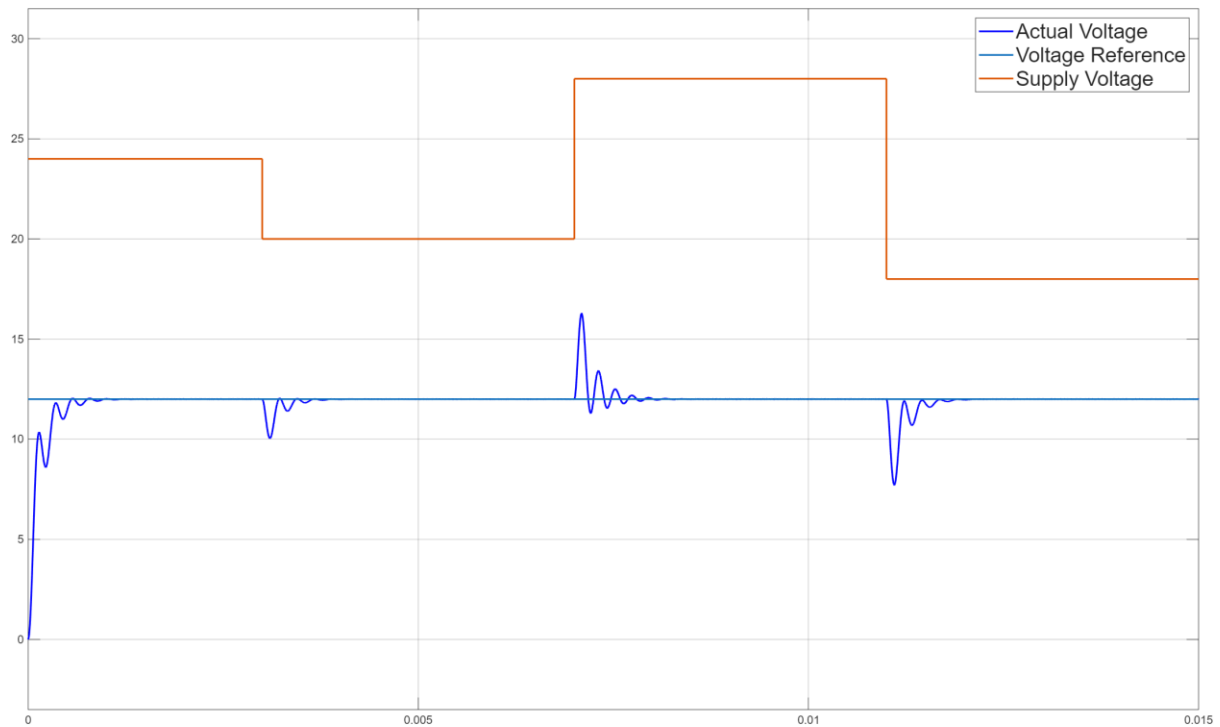


Figure 6.8 Line Regulation Response of the Closed-Loop Buck Converter

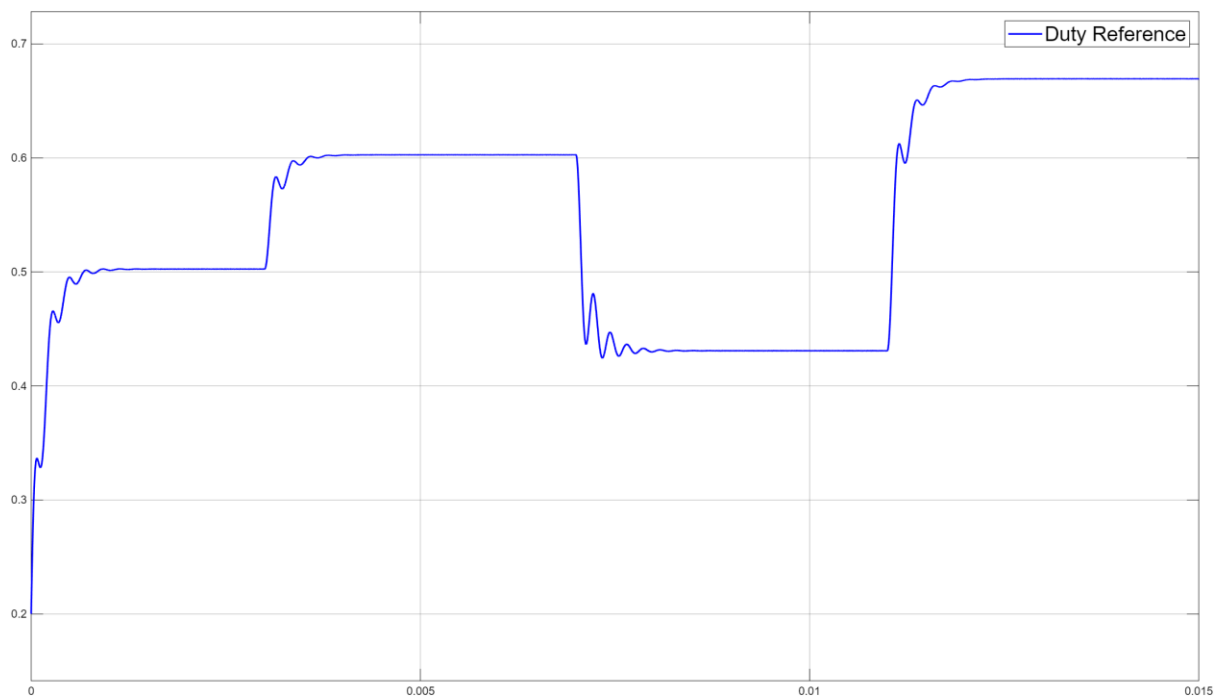


Figure 6.9 Duty Ratio Variation During Line Regulation

As shown in Figure 6.8, each change in the input supply voltage introduces a temporary disturbance in the output voltage. However, the PI controller rapidly compensates for these disturbances and restores the output voltage to the desired reference value of 12 V with only small transient deviations. This demonstrates

effective disturbance rejection and satisfactory voltage regulation under varying supply conditions.

The corresponding duty ratio variation is shown in Figure 6.9. When the input voltage decreases, the controller increases the duty ratio to maintain the desired output voltage. Conversely, when the input voltage increases, the duty ratio is reduced. After each transient, the duty ratio settles to a new steady-state value corresponding to the applied supply voltage, closely following the theoretical relationship

$$D = \frac{V_o}{V_{in}}$$

The observed duty ratios are also consistent with the expected values. For example, the duty ratio is approximately 0.5 at 24 V, increases to about 0.6 when the supply voltage is reduced to 20 V, decreases to approximately 0.43 for a 28 V input, and finally increases to around 0.67 when the supply voltage is reduced to 18 V. These values closely match the theoretical duty ratios required to maintain a regulated output voltage of 12 V.

From the small-signal control-to-output transfer function derived in Chapter 3, the input voltage appears in the numerator of the plant model. Consequently, changes in the supply voltage primarily modify the plant gain while leaving the pole locations essentially unchanged. The simulation results confirm that the designed PI controller effectively compensates for these gain variations, thereby maintaining stable operation and accurate voltage regulation over the tested range of input voltages.

6.5 Load Regulation

The load regulation capability of the closed-loop buck converter was evaluated by varying the load resistance while maintaining a constant input voltage of 24 V and a constant output voltage reference of 12 V. Unlike line regulation, load variations directly modify the converter operating point and alter the plant dynamics. Therefore, this test provides a more stringent evaluation of the designed PI controller.

The load variation was implemented in MATLAB/Simulink using a Variable Resistor block. The resistance value of the load was controlled by a Step signal through a switching arrangement, allowing discrete changes in the load resistance during simulation. During the test, the load resistance was varied from 6 Ω to 12

Ω , then to $2\ \Omega$, and finally returned to $6\ \Omega$, while the reference voltage remained fixed at $12\ \text{V}$ throughout the simulation.

The output voltage response obtained during the load regulation test is shown in Figure 6.10, the corresponding inductor current is shown in Figure 6.11, and the duty ratio generated by the PI controller is presented in Figure 6.12.

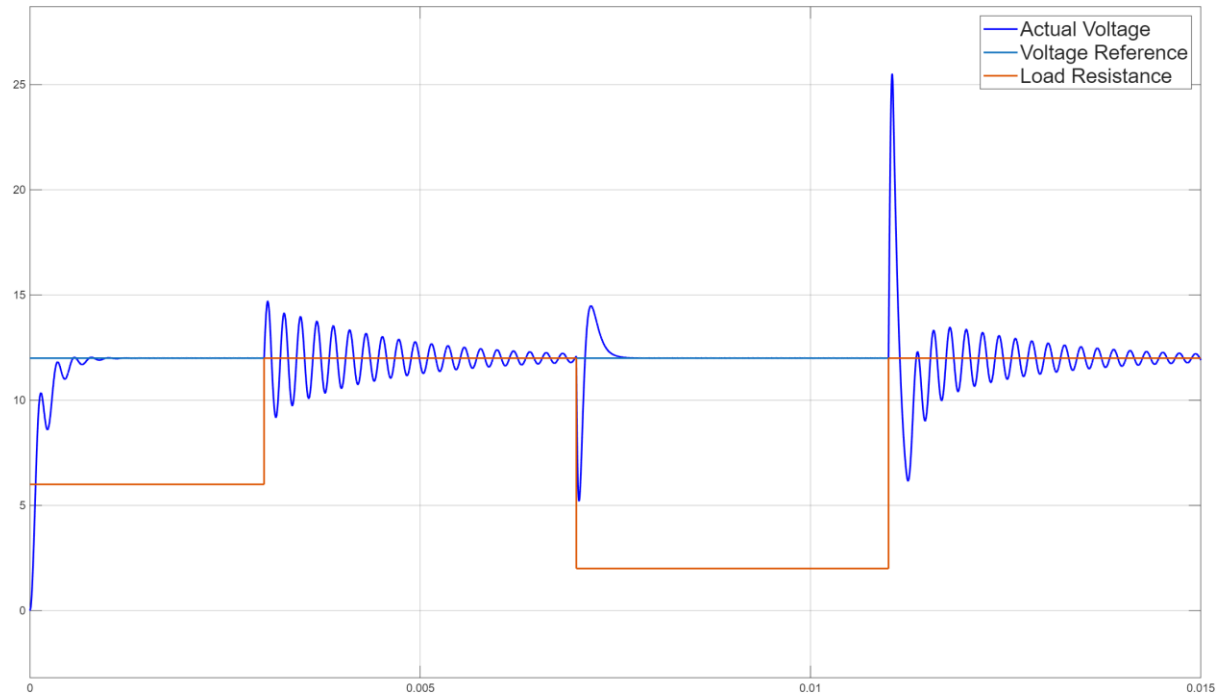


Figure 6.10 Load Regulation Response of the Closed-Loop Buck Converter

As shown in Figure 6.10, each change in the load resistance introduces a transient disturbance in the output voltage. When the load resistance is increased, the load current decreases, resulting in a temporary rise in the output voltage. Conversely, decreasing the load resistance increases the load current demand, producing a temporary voltage dip. In each case, the PI controller rapidly compensates for the disturbance and restores the output voltage to the desired reference value of $12\ \text{V}$, demonstrating satisfactory load regulation.

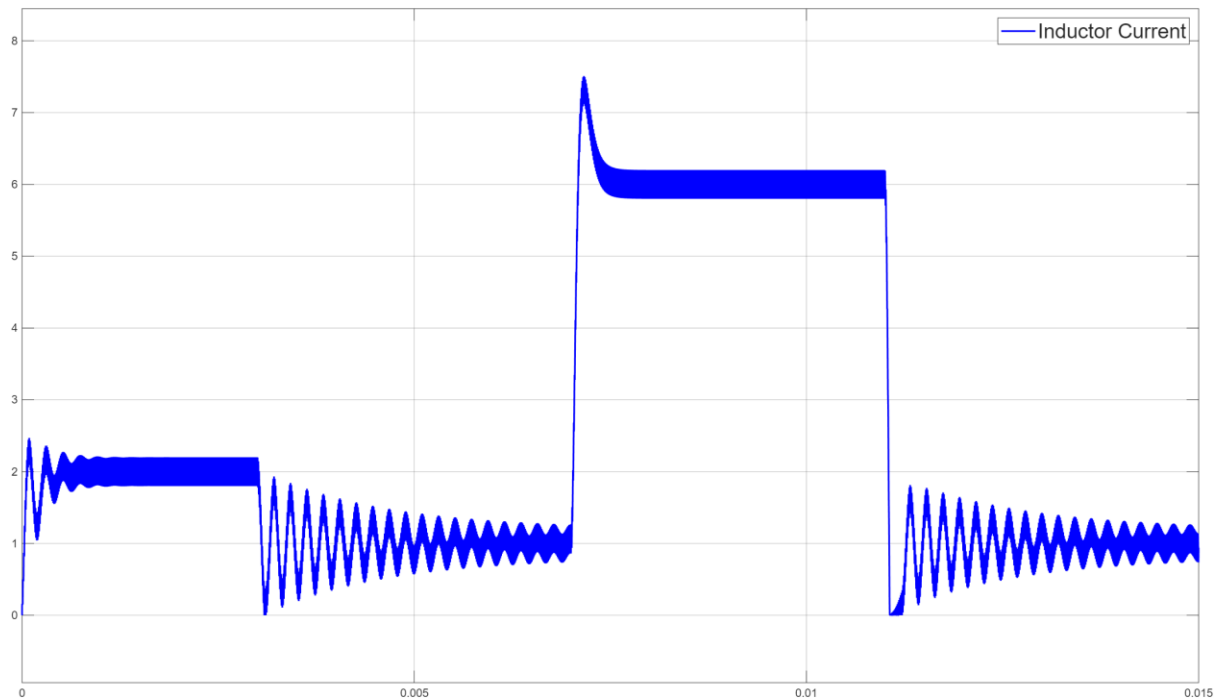


Figure 6.11 Inductor Current During Load Regulation

Figure 6.11 illustrates the corresponding inductor current response. As expected, the inductor current decreases when the load resistance is increased and increases significantly when the load resistance is reduced. Since the inductor current cannot change instantaneously, transient oscillations are observed immediately after each load change before settling to the new steady-state value required by the load.

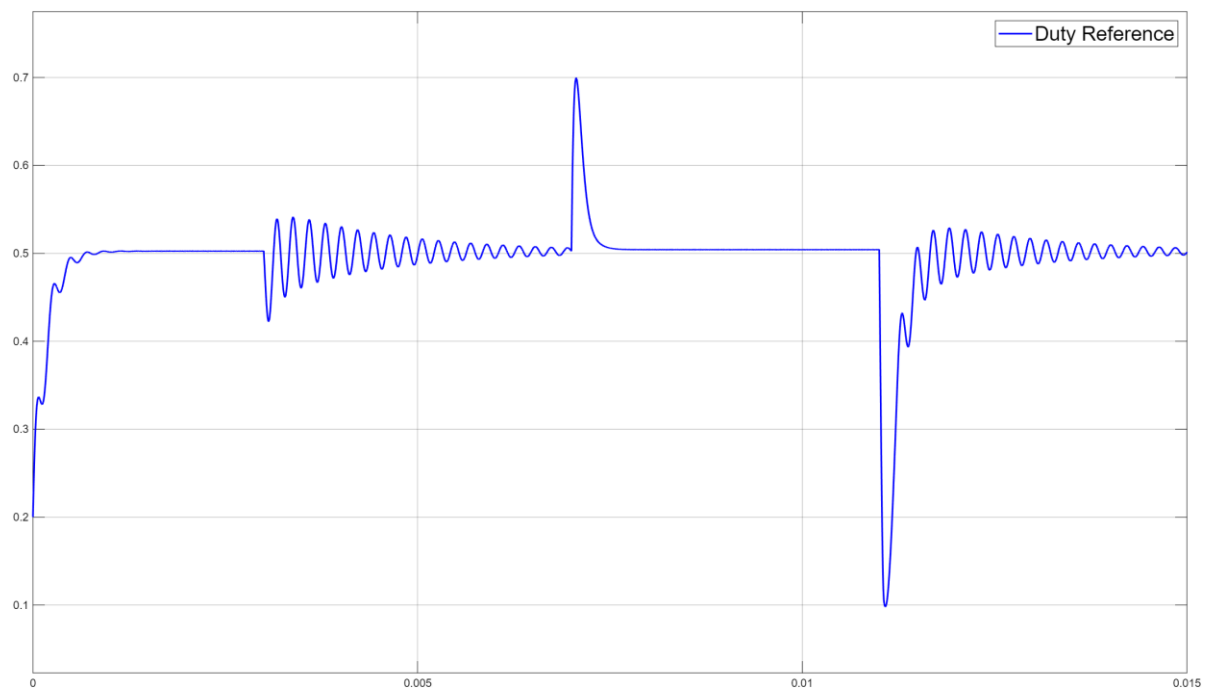


Figure 6.12 Duty Ratio Variation During Load Regulation

The duty ratio generated by the PI controller during the load variation test is shown in Figure 6.12. Following each load disturbance, the controller automatically adjusts the duty ratio to compensate for the resulting output voltage deviation. Once the transient response subsides, the duty ratio converges back to approximately its nominal operating value, since the input voltage and desired output voltage remain unchanged. These temporary duty ratio adjustments supply or remove the additional energy required during the transient interval, thereby restoring the output voltage to its reference value.

Unlike the line regulation test, load variation affects the converter transfer function through the load resistance term present in the denominator of the control-to-output transfer function. Consequently, the effective damping and dynamic response of the converter change with the load. Despite these variations in plant dynamics, the simulation results demonstrate that the designed PI controller maintains stable operation and provides effective voltage regulation over the tested load range.

7. Conclusion

This project presented the complete design, modeling, controller development, and validation of a closed-loop buck converter for regulated DC voltage conversion. The converter was first designed using analytical design equations, and the passive components were selected to ensure continuous conduction mode operation while satisfying the specified ripple requirements. The open-loop converter was implemented and validated using both MATLAB/Simulink and LTspice to verify the theoretical design.

A small-signal averaged model of the buck converter was then derived using the state-space averaging technique. The resulting control-to-output transfer function served as the mathematical model for controller design. Based on the desired transient performance specifications, a PI controller was designed using frequency-domain loop shaping techniques. The controller parameters were selected to achieve the required crossover frequency while maintaining adequate gain margin and phase margin for stable operation.

The designed controller was implemented in a closed-loop MATLAB/Simulink model using PWM-based duty cycle generation. The converter was subsequently evaluated under steady-state operation, static reference tracking, dynamic reference tracking, line regulation, and load regulation conditions. Simulation results demonstrated fast transient response, negligible steady-state error, satisfactory disturbance rejection, and stable operation over the entire operating range considered in this work. The obtained results confirm that the proposed controller successfully satisfies the design objectives and provides effective voltage regulation for the closed-loop buck converter.

Future Scope

The work presented in this project may be extended in several directions to further improve the converter performance and facilitate practical implementation.

- Hardware implementation of the complete closed-loop buck converter using a microcontroller or digital signal controller for real-time validation.
- PCB design and fabrication with appropriate gate driver circuitry, current sensing, and voltage feedback circuits.
- Investigation of advanced control techniques such as PID, lead-lag compensation, state-feedback control, sliding mode control, or model predictive control for improved dynamic performance and robustness.

- Inclusion of non-ideal component characteristics, parasitic resistances, switching losses, and thermal effects to obtain a more accurate converter model.
- Experimental evaluation under wider operating conditions, including larger input voltage variations, dynamic load profiles, and efficiency measurements.
- Extension of the developed control methodology to other DC–DC converter topologies such as the boost, buck-boost, SEPIC, and Ćuk converters.

This provides a natural progression from simulation-based controller design to practical implementation and advanced control strategies.

APPENDICES

Appendix A

MATLAB Code for Buck Converter Design Calculations

This appendix contains the MATLAB script used to calculate the design parameters of the buck converter presented in Chapter 2. The program computes the duty ratio, inductance and capacitance based on the specified converter ratings and ripple requirements.

```
%% =====  
% OPEN-LOOP BUCK CONVERTER ANALYSIS  
% =====  
  
clc;  
clearvars -except out;  
close all;  
  
%% Extract Data  
  
t = out.Vo.time;  
  
Vo = out.Vo.signals.values;  
Io = out.Io.signals.values;  
IL = out.I_L.signals.values;  
  
%% Select Steady-State Region (Last 20 ms)  
  
idx = t >= 0.18;  
  
t_ss = t(idx);  
  
Vo_ss = Vo(idx);  
Io_ss = Io(idx);  
IL_ss = IL(idx);  
  
%% Average Values  
  
Vo_avg = mean(Vo_ss);  
Io_avg = mean(Io_ss);  
IL_avg = mean(IL_ss);  
  
%% Peak-to-Peak Ripple  
  
Vo_ripple = max(Vo_ss) - min(Vo_ss);  
IL_ripple = max(IL_ss) - min(IL_ss);  
  
%% Voltage Ripple in mV  
  
Vo_ripple_mV = Vo_ripple*1000;  
  
%% RMS Values (Optional)  
  
Vo_rms = rms(Vo_ss);  
Io_rms = rms(Io_ss);
```

```

IL_rms = rms(IL_ss);

%% Display Results

fprintf('\n===== \n');
fprintf('          OPEN-LOOP BUCK CONVERTER RESULTS \n');
fprintf('===== \n');

fprintf('Average Output Voltage      : %.4f V \n',Vo_avg);
fprintf('Average Output Current        : %.4f A \n',Io_avg);
fprintf('Average Inductor Current       : %.4f A \n',IL_avg);

fprintf('\n');

fprintf('Output Voltage Ripple (p-p)    : %.2f mV \n',Vo_ripple_mV);
fprintf('Inductor Current Ripple (p-p)  : %.4f A \n',IL_ripple);

fprintf('\n');

fprintf('Output Voltage RMS             : %.4f V \n',Vo_rms);
fprintf('Output Current RMS             : %.4f A \n',Io_rms);
fprintf('Inductor Current RMS           : %.4f A \n',IL_rms);

fprintf('===== \n');

%% Plot Output Voltage

figure;
plot(t_ss,Vo_ss,'LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Output Voltage (V)');
title('Steady-State Output Voltage');

%% Plot Inductor Current

figure;
plot(t_ss,IL_ss,'LineWidth',1.5);
grid on;
xlabel('Time (s)');
ylabel('Inductor Current (A)');
title('Steady-State Inductor Current');

```

Appendix B

MATLAB/Simulink Model Parameters

This appendix summarizes the important simulation parameters and component characteristics used for implementing the closed-loop buck converter in MATLAB/Simulink. These settings were selected to ensure accurate representation of the converter switching behaviour and stable numerical simulation.

Table B.1 MATLAB/Simulink Simulation Parameters

Parameter	Value
Solver Type	Variable-step
Solver	ode23t (Modified Stiff/Trapezoidal)
Maximum Step Size	10^{-6} s
Relative Tolerance	10^{-3}

The maximum solver step size was limited to 1 μ s. Since the converter switching frequency is 100 kHz (switching period of 10 μ s), this setting provides approximately ten simulation points within each switching cycle, allowing accurate representation of the PWM waveform and switching transients.

Table B.2 Power Semiconductor Parameters

Component	Parameter	Value
MOSFET	$R_{DS(on)}$	0.01 Ω
Diode	Forward Resistance (R_f)	0.01 Ω
Diode	Forward Voltage	0.1 V

These semiconductor parameters were selected to model practical conduction losses while maintaining computational efficiency during simulation.

Appendix C

MATLAB Code for PI Controller Design

This appendix contains the MATLAB program used for the frequency-domain analysis and design of the PI controller presented in Chapter 4. The script defines the buck converter plant transfer function, implements the designed PI controller, generates the open-loop Bode plot, computes the gain margin, phase margin and crossover frequencies, and evaluates the closed-loop step response. The program also displays the plant, controller and closed-loop transfer functions used throughout the controller design process.

```
clc;
clear;
close all;

s = tf('s');

%% Motor Plant (Including PWM Gain)
G = 1.6e10/(s^2 + 16667*s + 666672400);

%% Controller
Kp = 0.016675;
Ki = 333.5;

C = Kp + Ki/s;

%% Open-Loop Transfer Function
L = C*G;

%% Bode Plot with Gain and Phase Margins
figure;
margin(L);
grid on;
title('Open-Loop Bode Plot with Gain and Phase Margin');

%% Gain Margin, Phase Margin and Crossover Frequencies
[GM, PM, Wcg, Wcp] = margin(L);

disp(' ');
disp('===== OPEN LOOP RESULTS =====');

if isinf(GM)
    disp('Gain Margin = Infinity');
else
    disp(['Gain Margin (dB) = ', num2str(20*log10(GM))]);
end

disp(['Phase Margin (deg) = ', num2str(PM)]);

disp(['Phase Crossover Frequency (rad/s) = ', num2str(Wcg)]);
disp(['Gain Crossover Frequency (rad/s) = ', num2str(Wcp)]);

%% Closed-Loop Transfer Function
```

```

T = feedback(L,1);

%% Closed-Loop Step Response
figure;
step(T);
grid on;
title('Closed-Loop Step Response');

%% Closed-Loop Performance Parameters
info = stepinfo(T);

disp(' ');
disp('==== CLOSED LOOP STEP RESPONSE ====');

disp(['Rise Time (s) = ', num2str(info.RiseTime)]);
disp(['Settling Time (s) = ', num2str(info.SettlingTime)]);
disp(['Overshoot (%) = ', num2str(info.Overshoot)]);
disp(['Peak Time (s) = ', num2str(info.PeakTime)]);
disp(['Peak Value = ', num2str(info.Peak)]);

%% Display Transfer Functions
disp(' ');
disp('==== PLANT TRANSFER FUNCTION ====');
G

disp(' ');
disp('==== CONTROLLER TRANSFER FUNCTION ====');
C

disp(' ');
disp('==== CLOSED LOOP TRANSFER FUNCTION ====');
T

```

Appendix D

MATLAB Code for Steady-State Performance Analysis

This appendix contains the MATLAB program used to evaluate the steady-state performance of the closed-loop buck converter under the rated operating condition of 24 V input, 12 V output reference, and 6 Ω load resistance. The script extracts the simulation data from MATLAB/Simulink and calculates the average output voltage, output current, inductor current, peak-to-peak voltage ripple, peak-to-peak inductor current ripple, RMS values, and the corresponding maximum and minimum values presented in Chapter 6.

```
%% =====  
% CLOSED-LOOP BUCK CONVERTER ANALYSIS  
% =====  
  
clc;  
clearvars -except out;  
close all;  
  
%% Extract Data  
  
t = out.Vo.time;  
  
Vo = out.Vo.signals.values;  
Io = out.Io.signals.values;  
IL = out.I_L.signals.values;  
  
%% Select Steady-State Region (Last 10% of Simulation)  
  
idx = t >= 0.9*t(end);  
  
t_ss = t(idx);  
  
Vo_ss = Vo(idx);  
Io_ss = Io(idx);  
IL_ss = IL(idx);  
  
%% Average Values  
  
Vo_avg = mean(Vo_ss);  
Io_avg = mean(Io_ss);  
IL_avg = mean(IL_ss);  
  
%% Peak-to-Peak Ripple  
  
Vo_ripple = max(Vo_ss) - min(Vo_ss);  
IL_ripple = max(IL_ss) - min(IL_ss);  
  
%% Voltage Ripple in mV  
  
Vo_ripple_mV = Vo_ripple*1000;  
  
%% RMS Values
```



```

Vo_rms = rms(Vo_ss);
Io_rms = rms(Io_ss);
IL_rms = rms(IL_ss);

%% Minimum and Maximum Values

Vo_max = max(Vo_ss);
Vo_min = min(Vo_ss);

IL_max = max(IL_ss);
IL_min = min(IL_ss);

%% Display Results

fprintf('\n===== \n');
fprintf('      CLOSED-LOOP BUCK CONVERTER RESULTS \n');
fprintf('===== \n');

fprintf('\nAverage Values \n');
fprintf('----- \n');
fprintf('Average Output Voltage      : %.4f V \n', Vo_avg);
fprintf('Average Output Current      : %.4f A \n', Io_avg);
fprintf('Average Inductor Current     : %.4f A \n', IL_avg);

fprintf('\nRipple Values \n');
fprintf('----- \n');
fprintf('Maximum Output Voltage      : %.4f V \n', Vo_max);
fprintf('Minimum Output Voltage      : %.4f V \n', Vo_min);
fprintf('Output Voltage Ripple (p-p) : %.2f mV \n', Vo_ripple_mV);

fprintf('\nMaximum Inductor Current    : %.4f A \n', IL_max);
fprintf('Minimum Inductor Current     : %.4f A \n', IL_min);
fprintf('Inductor Current Ripple (p-p) : %.4f A \n', IL_ripple);

fprintf('\nRMS Values \n');
fprintf('----- \n');
fprintf('Output Voltage RMS          : %.4f V \n', Vo_rms);
fprintf('Output Current RMS          : %.4f A \n', Io_rms);
fprintf('Inductor Current RMS        : %.4f A \n', IL_rms);

fprintf('===== \n');

```

Appendix E

MATLAB Code for Static Reference Tracking Analysis

This appendix contains the MATLAB program used to evaluate the static reference tracking performance of the closed-loop buck converter presented in Section 6.2. The script computes the steady-state output voltage, rise time, settling time, overshoot, and steady-state error for different reference voltages. The same program was used for all reference tracking tests by modifying only the reference voltage value.

```
clc;
clearvars -except out
close all;

%% Extract Output Voltage
t = out.Vo.time;
Vo = out.Vo.signals.values;

%% Reference Voltage (Constant)
ReferenceVoltage = out.Vref.signals.values;

%% Steady-State Output
N = 100; % Average last 100 samples
Vo_ss = mean(Vo(end-N+1:end));

%% Steady-State Error
Ess = abs(ReferenceVoltage - Vo_ss);
Ess_percent = (Ess/ReferenceVoltage)*100;

%% Step Response Characteristics
info = stepinfo(Vo,t,ReferenceVoltage);

%% Display Results
fprintf('\n===== \n');
fprintf('          CLOSED-LOOP BUCK CONVERTER \n');
fprintf('===== \n');
fprintf('Reference Voltage      : %.3f V \n',ReferenceVoltage);
fprintf('Steady-State Output    : %.3f V \n',Vo_ss);
fprintf('Rise Time              : %.6f s \n',info.RiseTime);
fprintf('Settling Time          : %.6f s \n',info.SettlingTime);
fprintf('Overshoot              : %.4f % \n',info.Overshoot);
fprintf('Steady-State Error     : %.5f V \n',Ess);
fprintf('Steady-State Error (%) : %.5f % \n',Ess_percent);
fprintf('===== \n');

%% Plot
figure;

plot(t,Vo,'b','LineWidth',2);
hold on;

yline(ReferenceVoltage,'r--','Reference','LineWidth',2);

grid on;
xlabel('Time (s)');
```

```
ylabel('Voltage (V)');  
title('Reference Tracking');  
legend('Output Voltage','Reference','Location','best');
```